5.1.14 Prove the validity of the limit

$$\lim_{x\to x_0}(ax+b) = ax_0+b$$

by converting to a statement about sequences.

5.1.15 Prove the validity of the limit $\lim_{x\to x_0} x^2 = x_0^2$ by converting to a statement about sequences.

5.1.16 Show that $\lim_{x\to 0} |x|/x$ does not exist by using the sequential definition of limit.
SEE NOTE 95

5.1.17 Prove that the limit $\lim_{x\to 0} \frac{1}{x}$ fails to exist by converting to a statement about sequences.

5.1.18 Prove that the limit $\lim_{x\to 0} \sin(1/x)$ fails to exist by converting to a statement about sequences.

5.1.19 Let x_0 be an accumulation point of the domain E of a function f. Prove that the limit $\lim_{x\to x_0} f(x)$ fails to exist if and only if there is a sequence of distinct points $\{e_n\}$ of E converging to x_0 but with $\{f(e_n)\}$ divergent.

5.1.20 Let f be the *characteristic function* of the rational numbers; that is, f is defined for all real numbers by setting $f(x) = 1$ if x is a rational number and $f(x) = 0$ if x is not a rational number. Determine where, if possible, the limit $\lim_{x\to x_0} f(x)$ exists.

5.1.21 Using the sequential definition, show that if $\lim_{x\to x_0} f(x) = L$, then

$$\lim_{x\to x_0} |f(x)| = |L|.$$

5.1.22 Find hypotheses under which you can prove that if $\lim_{x\to x_0} f(x) = L$, then

$$\lim_{x\to x_0} \sqrt{f(x)} = \sqrt{L}.$$

SEE NOTE 96

5.1.3 Limits (Mapping Definition)

✂ *Enrichment section. May be omitted.*

The essential idea behind a limit

$$\lim_{x\to x_0} f(x) = L$$

is that values of x close to x_0 get mapped by f into values close to L. We have been able to express this idea by using inequalities that express this closeness: δ-close for the x values and ε-close for the $f(x)$ values. This is essentially a mapping property that can be expressed by arbitrary open sets.

The following definition is equivalent to both Definitions 5.1 and 5.4.

Definition 5.6: (Limit) Let $f : E \to \mathbb{R}$ be a function with domain E and suppose that x_0 is a point of accumulation of E. Then we write

$$\lim_{x\to x_0} f(x) = L$$

if for every open set V containing the point L there is an open set U containing the point x_0 and every point $x \neq x_0$ of U that is in the domain of f is mapped into a point in V; that is,

$$f : E \cap U \setminus \{x_0\} \to V.$$

Once again, we must show that this definition is equivalent to the ε-δ definition. We prove that now.

Proof. **(Definitions 5.1 and 5.6 are equivalent)** Suppose first that

$$\lim_{x \to x_0} f(x) = L$$

according to Definition 5.1. Let V be an open set containing the point L. Then, since L is an interior point of V there is a positive number ε with

$$(L-\varepsilon, L+\varepsilon) \subset V.$$

Choose $\delta > 0$ so that

$$|f(x) - L| < \varepsilon$$

if $0 < |x - x_0| < \delta$ whenever x is a point in E (the domain of f). Let U be the open set $(x_0 - \delta, x_0 + \delta)$. Then the inequality we have shows that every point $x \neq x_0$ of U that is in the domain of f is mapped into a point in V. This is precisely Definition 5.6.

Conversely, suppose that $\lim_{x \to x_0} f(x) = L$ according to Definition 5.6. Let $\varepsilon > 0$. Choose $V = (L-\varepsilon, L+\varepsilon)$. By our definition there must be an open set U containing the point x_0 and every point $x \neq x_0$ of U that is in the domain of f is mapped into a point in V. Since x_0 is an interior point of U there must be a positive number δ so that

$$(x_0 - \delta, x_0 + \delta) \subset U.$$

This mapping property implies that

$$|f(x) - L| < \varepsilon$$

if $0 < |x - x_0| < \delta$. This is exactly our ε-δ definition of Definition 5.1. ■

Since all three of our definitions are equivalent we can use either a sequential argument, a mapping argument, or an ε-δ argument in our discussions of limits.

Exercises

5.1.23 Show that $\lim_{x \to 0} |x|/x$ does not exist using the mapping definition of limit.

5.1.24 Prove directly that the sequential definition of limit is equivalent to the mapping definition.

5.1.4 One-Sided Limits

It is possible for a function to fail to have a limit at a point and yet appear to have limits on one side. If we ignore what is happening on the right for a function, perhaps it will have a "left-hand limit." This is easy to achieve. Let f be defined everywhere near a point x_0 and define a new function

$$g(x) = f(x) \text{ for all } x < x_0.$$

This new function g is defined on a set to the left of x_0 and knows nothing of the values of f on the right. Thus the limit

$$\lim_{x \to x_0} g(x)$$

can be thought of as a left-hand limit for f. It would be written as

$$\lim_{x \to x_0-} f(x)$$

where the "x_0-" is the indication that a left-hand limit is used, not an ordinary limit. Similarly, the notation

$$\lim_{x \to x_0+} f(x)$$

denotes a right-hand limit with the "x_0+" indicating the limit on the positive or right side of x_0.

Since these one-sided limits are really just ordinary limits for a different function, they must satisfy all the theory of ordinary limits with no further fuss. We can use them quite freely without worrying that they need a different definition or a different theory. Even so, it is convenient to translate our usual definitions into one-sided limits just to have an expression for them. We give the right-hand version. You can supply a left-hand version.

Definition 5.7: (Right-Hand Limit) Let $f : E \to \mathbb{R}$ be a function with domain E and suppose that x_0 is a point of accumulation of $E \cap (x_0, \infty)$. Then we write

$$\lim_{x \to x_0+} f(x) = L$$

if for every $\varepsilon > 0$ there is a $\delta > 0$ so that

$$|f(x) - L| < \varepsilon$$

whenever $x_0 < x < x_0 + \delta$ and $x \in E$.

An equivalent sequential version can be established.

Definition 5.8: (Right-Hand Limit) Let $f : E \to \mathbb{R}$ be a function with domain E and suppose that x_0 is a point of accumulation of $E \cap (x_0, \infty)$. Then we write

$$\lim_{x \to x_0+} f(x) = L$$

if for every decreasing sequence $\{e_n\}$ of points of E with $e_n > x_0$ and $e_n \to x_0$ as $n \to \infty$,

$$\lim_{n \to \infty} f(e_n) = L.$$

Exercises

5.1.25 Show directly that Definitions 5.7 and 5.8 are equivalent.

5.1.26 Under appropriate additional assumptions about the domain of the function f show that $\lim_{x \to x_0} f(x) = L$ if and only if both

$$\lim_{x \to x_0+} f(x) = L \text{ and } \lim_{x \to x_0-} f(x) = L$$

are valid.

5.1.27 If the two limits

$$\lim_{x \to x_0+} f(x) = L_1 \text{ and } \lim_{x \to x_0-} f(x) = L_2$$

exist and are different, then the function is said to have a *jump discontinuity* at that point. The value $L_1 - L_2$ is called the *magnitude of the jump*. Give an example of a function with a jump of magnitude 3 at the value $x_0 = 2$. Give an example with a jump of magnitude -3.

5.1.28 Compute the one-sided limits of the function

$$f(x) = \frac{x}{|x|}$$

at any point x_0.

SEE NOTE 97

5.1.29 Compute, if possible, the one-sided limits of the function

$$f(x) = e^{1/x}$$

at 0.

SEE NOTE 98

5.1.30 According to our definitions, is there any distinction between the assertions

$$\lim_{x\to 0} \sqrt{x} = 0 \text{ and } \lim_{x\to 0+} \sqrt{x} = 0?$$

What is the meaning of $\lim_{x\to 0-} \sqrt{x} = 0$?

SEE NOTE 99

5.1.5 Infinite Limits

We can easily check that the limits

$$\lim_{x\to 0+} \frac{1}{x} \text{ and } \lim_{x\to 0-} \frac{1}{x}$$

fail to exist. A glance at the graph of the function $f(x) = 1/x$ suggests that we should write instead

$$\lim_{x\to 0+} \frac{1}{x} = \infty \text{ and } \lim_{x\to 0-} \frac{1}{x} = -\infty$$

as a way of conveying more information about what is happening rather than saying merely that the limits do not exist.

In this we are following our custom in the study of divergent sequences. Some sequences merely diverge, some diverge to ∞ or to $-\infty$. If we look back at the definition for sequences and compare it with our function limit definition, we should arrive at the following definition.

Definition 5.9: (Infinite Limit) Let $f : E \to \mathbb{R}$ be a function with domain E and suppose that x_0 is a point of accumulation of $E \cap (x_0, \infty)$. Then we write

$$\lim_{x\to x_0+} f(x) = \infty$$

if for every $M > 0$ there is a $\delta > 0$ so that $f(x) \geq M$ whenever

$$x_0 < x < x_0 + \delta \text{ and } x \in E.$$

Similarly, we can define

$$\lim_{x\to x_0+} f(x) = -\infty$$

if for every $m < 0$ there is a $\delta > 0$ so that $f(x) \leq m$ whenever $x_0 < x < x_0 + \delta$ and $x \in E$. The infinite limits on the left are similarly defined and denoted $\lim_{x\to x_0-} f(x) = \infty$ and $\lim_{x\to x_0-} f(x) = -\infty$. Also, two-sided limits are defined in the same manner, but with a two-sided condition.

Note. Just as for sequences, we do not say that the limit of a function *exists* unless that limit is finite. Thus, for example, we would say that the limit $\lim_{x\to 0+} 1/x$ does not exist, and that in fact $\lim_{x\to 0+} 1/x = \infty$. A limit is a real number. The symbols ∞ and $-\infty$ are used to describe certain situations, but they are not interpreted as numbers themselves.

Exercises

5.1.31 Give an equivalent formulation for infinite limits using a sequential version.

5.1.32 Formulate a definition for the statement that $\lim_{x\to x_0-} f(x) = \infty$. Using your definition, show that

$$\lim_{x\to x_0-} f(x) = \infty$$

if and only if

$$\lim_{x\to(-x_0)+} f(-x) = \infty.$$

5.1.33 Where does the function

$$f(x) = \frac{1}{\sqrt{x^2-1}}$$

have infinite limits? Give proofs using the definition.

5.1.34 Formulate a definition for the statements

$$\lim_{x\to\infty} f(x) = L \text{ and } \lim_{x\to-\infty} f(x) = L.$$

SEE NOTE 100

5.1.35 Formulate a definition for the statements

$$\lim_{x\to\infty} f(x) = \infty \text{ and } \lim_{x\to-\infty} f(x) = \infty.$$

5.1.36 Let $f : (0,\infty) \to \mathbb{R}$. Show that

$$\lim_{x\to\infty} f(x) = L \text{ if and only if } \lim_{x\to 0+} f(1/x) = L.$$

5.1.37 What are the limits $\lim_{x\to\infty} x^p$ for various real numbers p?

5.1.38 Show that one of the limits $\lim_{x\to 0+} f(x)$ and $\lim_{x\to 0-} f(x)$ of the function

$$f(x) = e^{1/x}$$

at 0 is infinite and one is finite. What can you say about the limits

$$\lim_{x\to\infty} f(x) \text{ and } \lim_{x\to-\infty} f(x)?$$

SEE NOTE 101

5.2 Properties of Limits

The computation of limits in calculus courses depended on a theory of limits. For most simple computations it was enough to know how to handle functions that were put together by adding, subtracting, multiplying, or dividing other functions. Later, more subtle problems required advanced techniques (e.g., L'Hôpital's rule). Here we develop the rudiments of a theory of function limits.

We start with the uniqueness property, the boundedness property and continue to the algebraic properties. In this we are following much the same path we did when we began our study of sequential limits. Indeed the definitions of sequential limits and function limits are so similar that the theories are necessarily themselves quite similar.

5.2.1 Uniqueness of Limits

When we write the statement

$$\lim_{x \to x_0} f(x) = L$$

we wish to be assured that it is not also true for some other numbers different from L.

Theorem 5.10 (Uniqueness of Limits) *Suppose that*

$$\lim_{x \to x_0} f(x) = L.$$

Then the number L is unique: No other number has this same property.

Proof. We suppose that

$$\lim_{x \to x_0} f(x) = L$$

and

$$\lim_{x \to x_0} f(x) = L_1$$

are both true. To prove the theorem we must show that $L = L_1$. If we convert this to a statement about sequences this asserts that any sequence $x_n \to x_0$ with $x_n \neq x_0$ and all points in the domain of f must have

$$f(x_n) \to L$$

and also must have

$$f(x_n) \to L_1.$$

For these limits to exist the point x_0 must be a point of accumulation for the domain of f and so there exists at least one such sequence. But we have already established for sequence limits that this is impossible (Theorem 2.8) unless $L = L_1$. ■

Exercises

5.2.1 Give an ε-δ proof of Theorem 5.10.
SEE NOTE 102

5.2.2 Explain why the proof fails if the part of the limit definition that asserts x_0 is to be a point of accumulation of the domain of f were omitted.

5.2.2 Boundedness of Limits

We recall that convergent sequences are bounded. There is a similar statement for functions. If a function limit exists the function cannot be too large; the statement must be made precise, however, since it is really only valid close to the point where the limit is taken.

For example, you will recall from our discussion of local boundedness in Section 4.5 that the function $f(x) = 1/x$ is unbounded and yet locally bounded at each point other than at 0. In the same way we will see that the existence of the limit

$$\lim_{x \to x_0} \frac{1}{x} = \frac{1}{x_0}$$

for every value of $x_0 \neq 0$ also requires that local boundedness property.

Theorem 5.11 (Boundedness of Limits) *Suppose that the limit*

$$\lim_{x \to x_0} f(x) = L$$

exists. Then there is an interval $(x_0 - c, x_0 + c)$ and a number M such that

$$|f(x)| \leq M$$

for every value of x in that interval that is in the domain of f.

Proof. There is a $\delta > 0$ so that

$$|f(x) - L| < 1$$

whenever x is a point in the domain of f differing from x_0 and satisfying $|x - x_0| < \delta$. If x_0 is not in the domain of f, then this means that

$$|f(x)| = |f(x) - L + L| \leq |f(x) - L| + |L| < |L| + 1$$

for all x in $(x_0 - \delta, x_0 + \delta)$ that are in the domain of f. This would complete the proof since we can take $M = |L| + 1$.

If x_0 is in the domain of f, then take instead

$$M = |L| + 1 + |f(x_0)|.$$

Then

$$|f(x)| \leq M$$

for all x in $(x_0 - \delta, x_0 + \delta)$ that are in the domain of f. ■

A similar statement can be made about boundedness away from zero. This shows that if a function has a nonzero limit, then close by to the point the function stays away from zero. The proof uses similar ideas and is left for the exercises.

Theorem 5.12 (Boundedness Away from Zero) *If the limit*

$$\lim_{x \to x_0} f(x)$$

exists and is not zero, then there is an interval $(x_0 - c, x_0 + c)$ *and a positive number* m *such that*

$$|f(x)| \geq m > 0$$

for every value of $x \neq x_0$ *in that interval and that belongs to the domain of* f.

Exercises

5.2.3 Prove Theorem 5.11 using the sequential definition of limit instead.
SEE NOTE 103

5.2.4 Use Theorem 5.11 to show that $\lim_{x\to 0} \frac{1}{x}$ cannot exist.

5.2.5 Prove Theorem 5.12 using an ε-δ argument.

5.2.6 Prove Theorem 5.12 using a sequential argument.

5.2.7 Prove Theorem 5.12 by deriving it from Theorem 5.11 and the fact (proved later) that if

$$\lim_{x\to x_0} f(x) = L \neq 0$$

then

$$\lim_{x\to x_0} \frac{1}{f(x)} = \frac{1}{L}.$$

5.2.3 Algebra of Limits

Functions can be combined by the usual arithmetic operations (addition, subtraction, multiplication and division). Indeed most functions we are likely to have encountered in a calculus course can be seen to be composed of simpler functions combined together in this way.

Example 5.13: The computations

$$\lim_{x\to 3} \frac{2x^3+4}{3x^2+1} = \frac{\lim_{x\to 3}(2x^3+4)}{\lim_{x\to 3}(3x^2+1)}$$

$$= \frac{2(\lim_{x\to 3} x^3)+4}{3(\lim_{x\to 3} x^2)+1} = \frac{2\times 3^3+4}{3\times 3^2+1}$$

should return fond memories of calculus homework assignments. But how are these computations properly justified? ◀

Because of our experience with sequence limits, we can anticipate that there should be an "algebra of function limits" just as there was an algebra of sequence limits. The proofs can be obtained either by imitating the proofs we constructed earlier for sequences or by using the fact that function limits can be reduced to sequential limits.

There is an extra caution here. An example illustrates.

Example 5.14: We know that $\lim_{x\to 0} \sqrt{-x} = 0$ and $\lim_{x\to 0} \sqrt{x} = 0$. Does it follow that

$$\lim_{x\to 0} \left(\sqrt{x} + \sqrt{-x}\right) = 0?$$

There is only one point in the domain of the function

$$f(x) = \sqrt{x} + \sqrt{-x}$$

and so no limit statement is possible. ◀

The extra hypothesis throughout the following theorems appears in order to avoid examples like this. We must assume that the domain of f, call it $\text{dom}(f)$, and the domain of g, call it $\text{dom}(g)$, must have enough points in common to define the limit at the point x_0 being considered. In most simple applications the domains of the functions do not cause any troubles.

For proofs we have a number of strategies available. We can reduce these limit theorems to statements about sequences and then appeal to the theory of sequential limits that we developed in Chapter 2. Alternatively, we can construct ε-δ proofs by modeling them after the similar statements that we proved for sequences. We do not need any really new ideas. The proofs have, accordingly, been left to the exercises.

Theorem 5.15 (Multiples of Limits) *Suppose that the limit*

$$\lim_{x \to x_0} f(x)$$

exists and that C is a real number. Then

$$\lim_{x \to x_0} Cf(x) = C\left(\lim_{x \to x_0} f(x)\right).$$

Theorem 5.16 (Sums and Differences) *Suppose that the limits*

$$\lim_{x \to x_0} f(x) \text{ and } \lim_{x \to x_0} g(x)$$

exist and that x_0 is a point of accumulation of $dom(f) \cap dom(g)$. Then

$$\lim_{x \to x_0} (f(x) + g(x)) = \lim_{x \to x_0} f(x) + \lim_{x \to x_0} g(x)$$

and

$$\lim_{x \to x_0} (f(x) - g(x)) = \lim_{x \to x_0} f(x) - \lim_{x \to x_0} g(x).$$

Theorem 5.17 (Products of Limits) *Suppose that the limits*

$$\lim_{x \to x_0} f(x) \text{ and } \lim_{x \to x_0} g(x)$$

exist and that x_0 is a point of accumulation of $dom(f) \cap dom(g)$. Then

$$\lim_{x \to x_0} f(x)g(x) = \left(\lim_{x \to x_0} f(x)\right)\left(\lim_{x \to x_0} g(x)\right).$$

Theorem 5.18 (Quotients of Limits) *Suppose that the limits*

$$\lim_{x \to x_0} f(x) \text{ and } \lim_{x \to x_0} g(x)$$

exist and that the latter is not zero and that x_0 is a point of accumulation of $dom(f) \cap dom(g)$. Then

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x \to x_0} f(x)}{\lim_{x \to x_0} g(x)}.$$

Exercises

5.2.8 Let f and g be functions with domains $\text{dom}(f)$ and $\text{dom}(g)$. What are the domains of the functions listed below obtained by combining these functions algebraically or by a composition?

(a) $f+g$

(b) $f-g$

(c) fg

(d) f/g

(e) $f \circ g$

(f) $\sqrt{f+g}$

(g) $\sqrt{fg}$

5.2.9 What exactly is the trouble that arises in the theorems of this section that required us to assume "that x_0 is a point of accumulation of $\text{dom}(f) \cap \text{dom}(g)$?"
SEE NOTE 104

5.2.10 Is it true that if both $\lim_{x\to x_0} f(x)$ and $\lim_{x\to x_0} g(x)$ fail to exist, then

$$\lim_{x\to x_0} (f(x)+g(x))$$

must also fail to exist?

5.2.11 In the statement of Theorem 5.18 don't we also have to assume that $g(x)$ is never zero?
SEE NOTE 105

5.2.12 A careless student gives the following as a proof of Theorem 5.17. Find the flaw: "Suppose that $\varepsilon > 0$. Choose δ_1 so that

$$|f(x)-L| < \frac{\varepsilon}{2|M|+1}$$

if $0 < |x-x_0| < \delta_1$ and also choose δ_2 so that

$$|g(x)-M| < \frac{\varepsilon}{2|f(x)|+1}$$

if $0 < |x-x_0| < \delta_2$ Define $\delta = \min\{\delta_1, \delta_2\}$. If $0 < |x-x_0| < \delta$, then we have

$$\begin{aligned} |f(x)g(x)-LM| &\le |f(x)|\,|g(x)-M| + |M|\,|f(x)-L| \\ &\le |f(x)|\left(\frac{\varepsilon}{2|f(x)|+1}\right) + |M|\left(\frac{\varepsilon}{2|M|+1}\right) < \varepsilon. \end{aligned}$$

Well, that shows $f(x)g(x) \to LM$ if $f(x) \to L$ and $g(x) \to M$."

5.2.13 Prove Theorem 5.15 by using an ε-δ proof and by using the sequential definition of limit.

5.2.14 Prove Theorem 5.16 by using an ε-δ proof and by using the sequential definition of limit.

5.2.15 Prove Theorem 5.18 by using the sequential definition of limit.

5.2.16 Prove Theorem 5.17 by correcting the flawed ε-δ proof in Exercise 5.2.12 and by using the sequential definition of limit. Which method is easier?

.

5.2.4 Order Properties

Just as we saw that sequence limits preserve both the algebraic structure and the order structure, so we will find that function limits have the same properties. We have just completed the algebraic properties. We turn now to the order properties.

If $f(x) \leq g(x)$ for all x, then we expect to conclude that

$$\lim_{x \to x_0} f(x) \leq \lim_{x \to x_0} g(x).$$

We now prove this and several other properties that relate directly to the order structure of the real numbers.

Theorem 5.19: *Suppose that the limits*

$$\lim_{x \to x_0} f(x) \quad \textit{and} \quad \lim_{x \to x_0} g(x)$$

exist and that x_0 is a point of accumulation of $dom(f) \cap dom(g)$. If

$$f(x) \leq g(x)$$

for all $x \in dom(f) \cap dom(g)$, then

$$\lim_{x \to x_0} f(x) \leq \lim_{x \to x_0} g(x).$$

Proof. Let us give an indirect proof. Let

$$L = \lim_{x \to x_0} f(x) \text{ and } M = \lim_{x \to x_0} g(x)$$

and suppose, contrary to the theorem, that $L > M$. Choose ε so small that $M + \varepsilon < L - \varepsilon$; that is, choose

$$\varepsilon < (L - M)/2.$$

By the definition of limits there are numbers δ_1 and δ_2 so that

$$f(x) > L - \varepsilon$$

if $x \neq x_0$ is within δ_1 of x_0 and in the domain of f and

$$g(x) < M + \varepsilon$$

if $x \neq x_0$ is within δ_2 of x_0 and is in the domain of g. But the conditions in the theorem assure us that there must be at least one point, $x = z$ say, that satisfies both conditions. That would mean

$$g(z) < M + \varepsilon < L - \varepsilon < f(z).$$

This is impossible as it contradicts the fact that all the values of $f(x)$ are less than the values $g(x)$. This contradiction completes the proof. ■

Note. There is a trap here that we encountered in our discussions of sequence limits. We remember that the condition $s_n < t_n$ does not imply that

$$\lim_{n \to \infty} s_n < \lim_{n \to \infty} t_n.$$

In the same way the condition $f(x) < g(x)$ does not imply

$$\lim_{x\to x_0} f(x) < \lim_{x\to x_0} g(x).$$

Be careful with this, too.

Corollary 5.20: *Suppose that the limit*

$$\lim_{x\to x_0} f(x)$$

exists and that $\alpha \le f(x) \le \beta$ *for all x in the domain of f. Then*

$$\alpha \le \lim_{x\to x_0} f(x) \le \beta.$$

Note. Again, don't forget the trap. The condition $\alpha < f(x) < \beta$ for all x implies at best that

$$\alpha \le \lim_{x\to x_0} f(x) \le \beta.$$

It would not imply that

$$\alpha < \lim_{x\to x_0} f(x) < \beta.$$

The next theorem is another useful variant on these themes. Here an unknown function is sandwiched between two functions whose limit behavior is known, allowing us to conclude that a limit exists. This theorem is often taught as "the squeeze theorem" just as the version for sequences in Theorem 2.20 was labeled. Here we need the functions to have the same domain.

Theorem 5.21 (Squeeze Theorem) *Suppose that* $f, g, h : E \to \mathbb{R}$ *and that* x_0 *is a point of accumulation of the common domain E. Suppose that the limits*

$$\lim_{x\to x_0} f(x) = L \text{ and } \lim_{x\to x_0} g(x) = L$$

exist and that

$$f(x) \le h(x) \le g(x)$$

for all $x \in E$ *except perhaps at* $x = x_0$. *Then* $\lim_{x\to x_0} h(x) = L$.

Proof. The easiest proof is to use a sequential argument. This is left as Exercise 5.2.19. ■

Example 5.22: Let us prove that the limit

$$\lim_{x\to 0} x\sin(1/x) = 0$$

is valid. Certainly the expression $\sin(1/x)$ seems troublesome at first. But we notice that the inequalities

$$-|x| \le x\sin(1/x) \le |x|$$

are valid for all x (except $x = 0$ where the function is undefined). Since

$$\lim_{x\to 0} |x| = \lim_{x\to 0} -|x| = 0$$

Theorem 5.21 supplies our result. ◄

A final theorem on the theme of order structure is often needed. The absolute value, we recall, is defined directly in terms of the order structure. Is absolute value preserved by the limit operation? As the proof does not require any new ideas, it is left as Exercise 5.2.21.

Theorem 5.23 (Limits of Absolute Values) *Suppose that the limit*

$$\lim_{x\to x_0} f(x) = L$$

exists. Then

$$\lim_{x\to x_0} |f(x)| = |L|.$$

Since maxima and minima can be expressed in terms of absolute values, there is a corollary that is sometimes useful.

Corollary 5.24 (Max/Min of Limits) *Suppose that the limits*

$$\lim_{x\to x_0} f(x) = L \text{ and } \lim_{x\to x_0} g(x) = M$$

exist and that x_0 is a point of accumulation of $dom(f) \cap dom(g)$. Then

$$\lim_{x\to x_0} \max\{f(x), g(x)\} = \max\{L, M\}$$

and

$$\lim_{x\to x_0} \min\{f(x), g(x)\} = \min\{L, M\}.$$

Proof. The first of the these follows from the identity

$$\max\{f(x), g(x)\} = \frac{f(x)+g(x)}{2} + \frac{|f(x)-g(x)|}{2}$$

and the theorem on limits of sums and the theorem on limits of absolute values. In the same way the second assertion follows from

$$\min\{f(x), g(x)\} = \frac{f(x)+g(x)}{2} - \frac{|f(x)-g(x)|}{2}.$$

■

Exercises

5.2.17 Show that the condition $f(x) < g(x)$ does not imply that

$$\lim_{x\to x_0} f(x) < \lim_{x\to x_0} g(x).$$

5.2.18 Give a sequential type proof for Theorem 5.19.

5.2.19 Give a sequential type proof for Theorem 5.21.

5.2.20 Give an ε-δ proof of Theorem 5.23.

5.2.21 Give a proof of Theorem 5.23 by converting it to a statement about sequences.

5.2.22 Extend Corollary 5.24 to the case of more than two functions; that is, determine

$$\lim_{x\to x_0} \max\{f_1(x), f_2(x), \ldots, f_n(x)\}.$$

5.2.5 Composition of Functions

You will have observed a pattern that is attractive in the study of limits. These examples suggest the pattern:

$$\lim_{x\to x_0} [f(x)]^2 = \left(\lim_{x\to x_0} f(x)\right)^2,$$

$$\lim_{x\to x_0} \sqrt{f(x)} = \sqrt{\lim_{x\to x_0} f(x)},$$

$$\lim_{x\to x_0} e^{f(x)} = e^{\lim_{x\to x_0} f(x)}.$$

The first is easy to prove since $[f(x)]^2 = f(x)f(x)$ and we can use the product rule. The square root example is harder but could be proved using an ε-δ argument and requires only the assumption that $\lim_{x\to x_0} f(x)$ is positive. It could be false if $\lim_{x\to x_0} f(x) = 0$ and definitely is false if $\lim_{x\to x_0} f(x) < 0$.

The third will require some familiarity with the exponential function and is harder still, though always true.

The general pattern is the following. Some function F is composed with f, and the limit computation we wish to use is

$$\lim_{x\to x_0} F(f(x)) = F\left(\lim_{x\to x_0} f(x)\right).$$

Can this be justified? More correctly, what are the conditions under which it can be justified?

Let us analyze this using a sequence argument since that often simplifies function limits. We suppose $x_n \to x_0$. We have then our supposition that $f(x_n) \to L$. Can we conclude

$$F(f(x_n)) \to F(L)?$$

This is exactly what we are doing when we try to use

$$\lim_{x\to x_0} F(f(x)) = F\left(\lim_{x\to x_0} f(x)\right).$$

The property of the function F that we desire is simple:

$$\textit{If } z_n \to z_0 \textit{ then } F(z_n) \to F(z_0).$$

Think of $z_n = f(x_n)$; then $z_n \to L$ and the required property is

$$F(z_n) \to F(L) \text{ whenever } z_n \to L.$$

This is the same as requiring that

$$\lim_{z\to L} F(z) = F(L).$$

Thus we have proved the following theorem, which completely answers our question about justifying the preceding operations.

Theorem 5.25: *Let F be a function defined in a neighborhood of the point L and such that*

$$\lim_{z\to L} F(z) = F(L).$$

If

$$\lim_{x\to x_0} f(x) = L$$

then

$$\lim_{x\to x_0} F(f(x)) = F\left(\lim_{x\to x_0} f(x)\right) = F(L).$$

The condition on the function F that

$$\lim_{z\to L} F(z) = F(L)$$

is called *continuity at the point L* and is the subject of Section 5.4.

Exercises

5.2.23 Show that

$$\lim_{x\to x_0} \sqrt{f(x)} = \sqrt{\lim_{x\to x_0} f(x)}$$

could be false if $\lim_{x\to x_0} f(x) = 0$ and definitely is false if $\lim_{x\to x_0} f(x) < 0$.

5.2.24 Give a formal proof of Theorem 5.25 using the sequential method sketched in the text.

5.2.25 Give a formal proof of Theorem 5.25 using an ε-δ method.

5.2.26 Give a formal proof of Theorem 5.25 using the mapping idea.

5.2.27 Give an example of a limit for which

$$\lim_{x\to x_0} F(f(x)) \neq F\left(\lim_{x\to x_0} f(x)\right)$$

even though both of the limits in the statement do exist.

5.2.28 Show that

$$\lim_{x\to x_0} |f(x)| = \left|\lim_{x\to x_0} f(x)\right|$$

under some appropriate assumption by applying Theorem 5.25.
SEE NOTE 106

5.2.29 Show that

$$\lim_{x\to x_0} \sqrt{|f(x)|} = \sqrt{\left|\lim_{x\to x_0} f(x)\right|}$$

under some appropriate assumption by applying Theorem 5.25.
SEE NOTE 107

5.2.30 Obtain Corollary 5.24 as an application of Theorem 5.25.

5.2.6 Examples

There are a number of well-known examples of limits that every student should know. Partly this is because there will be an expectation in later courses that these should have been seen. But, more important, an abundance of examples is needed to gain some insight into when limits exist and when they do not and how they behave.

For any function f defined near a point x_0 there are several possibilities we should look for.

1. Does the limit $\lim_{x\to x_0} f(x)$ exist?
2. If the limit does exist, is the limit the most likely value, namely
$$\lim_{x\to x_0} f(x) = f(x_0)?$$
(Such functions are said to be *continuous* at the point x_0.)
3. If the limit fails to exist, then could it be that the one-sided limits do exist but happen to be unequal; that is,
$$\lim_{x\to x_0+} f(x) \neq \lim_{x\to x_0-} f(x)?$$
(Such a function is said to have a *jump discontinuity* at the point x_0.)

The case that is most familiar, namely where
$$\lim_{x\to x_0} f(x) = f(x_0),$$
is described by the language of continuity. Our study of continuity comes in the next section. But let us be aware now of when a function has this property.

Polynomials All polynomial functions have entirely predictable limits. If
$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n,$$
then
$$\lim_{x\to x_0} p(x) = p(x_0)$$
at every value. (In the language we shall use, these functions are continuous.) To prove this we can use the fact that $\lim_{x\to x_0} a_0 = a_0$ and the fact that $\lim_{x\to x_0} x = x_0$. These are trivial to prove. Then the polynomial is built up from this by additions and multiplications. The theorems of Section 5.2.3 can be used to complete the verification [e.g., $\lim_{x\to x_0} x^2 = x_0^2$ by the product rule, $\lim_{x\to x_0} x^3 = \lim_{x\to x_0}(x)(x^2) = x_0^3$ by the product rule applied again].

Rational Functions A rational function is a function of the form
$$R(x) = \frac{p(x)}{q(x)}$$
where p and q are polynomials (i.e., a ratio of polynomials and hence the name). Since we can take limits
$$\lim_{x\to x_0} R(x) = \frac{\lim_{x\to x_0} p(x)}{\lim_{x\to x_0} q(x)}$$

freely, excepting only the case where the denominator is zero, we have found that

$$\lim_{x\to x_0} R(x) = R(x_0)$$

except at those points where $q(x_0) = 0$. At those points, it is possible that the limit exists. Note, however, that it cannot equal $R(x_0)$ since $R(x_0)$ is not defined. It is also possible that the right-hand and left-hand limits are infinite. There are some examples in the exercises to illustrate these possibilities.

Exponential Functions The exponential function e^x can be proved to have the limiting value that we would expect, namely

$$\lim_{x\to x_0} e^x = e^{x_0}.$$

To prove this depends on how we have defined the exponential function in the first place. There are many ways in which we can develop such a theory. It is usual to wait for more theoretical apparatus and then define the exponential function in an appropriate way that allows that to be exploited. Recall that we mentioned in Example 2.36 that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \ldots.$$

Sums like this are called power series. As part of the theory of power series we will discover precisely when they are continuous. Then it is possible to define the exponential function as a power series and claim continuity immediately.

Most of the elementary functions of the calculus (trigonometric functions, inverse trigonometric functions, etc.) can be handled in this way. We do not pause here to worry about limits of such functions.

Characteristic Function of the Rationals The *characteristic function* of a set E of real numbers is the function that assigns value 1 at points in E and value 0 at points outside E. Some authors call it an *indicator function* since it does, indeed, indicate when points are or are not in the set. For an interesting example of a function that would have been considered bizarre in the early days of calculus, consider the characteristic function of the rationals:

$$\chi_{\mathbb{Q}}(x) = 1 \text{ if } x \in \mathbb{Q}$$

and

$$\chi_{\mathbb{Q}}(x) = 0 \text{ if } x \notin \mathbb{Q}.$$

It is an easy exercise to check that

$$\lim_{x\to x_0+} \chi_{\mathbb{Q}}(x) \text{ and } \lim_{x\to x_0-} \chi_{\mathbb{Q}}(x)$$

both fail to exist.

Dirichlet Function The Dirichlet function is defined on $[0,1]$ by

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is irrational or } x = 0 \\ 1/q, & \text{if } x = p/q,\ p,q \in \mathbb{N},\ p/q \text{ in lowest terms.} \end{cases}$$

To examine the limiting behavior of this function, we need to observe that while there are many points where this function is positive (all rationals) there are not many points

where it assumes a value greater than some positive number ε. Indeed if we count them we will see that for any positive integer q the set of points

$$S_q = \{x \in [0,1] : f(x) \geq 1/q\}$$

contains at most $q(q-1)/2$ points. The exact number is not important; all we need to observe is that there are only finitely many such points.

Thus let $\varepsilon > 0$ and choose any integer q large enough so that $1/q < \varepsilon$. For any point x_0 we can choose $\delta > 0$ in such a way that both intervals $(x_0 - \delta, x_0)$ and $(x_0, x_0 + \delta)$ contain no points of the finite set S_q. That must mean that every point x in $(x_0 - \delta, x_0)$ or $(x_0, x_0 + \delta)$ satisfies

$$0 \leq f(x) < 1/q < \varepsilon.$$

Thus it follows that

$$\lim_{x \to x_0} f(x) = 0$$

at every point x_0. In particular, the equation

$$\lim_{x \to x_0} f(x) = f(x_0)$$

will hold at every irrational point x_0 but must fail at every rational point. In the language of continuity we have proved that this function is continuous at every irrational point but discontinuous at every rational point. A curious function: It appears to be continuous at nearly every point and discontinuous at nearly every point. Nineteenth-century mathematicians were quite intrigued by such functions and called them pointwise discontinuous, a term that seems not to have survived.

(We shall return to this example occasionally. For example, Exercise 7.5.4 asks for an account of the local extrema of this function.)

Nondecreasing Functions with Jumps The simplest example of a function with a discontinuity is perhaps

$$H(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

This function fails to have a limit at $x = 0$ since $\lim_{x \to 0+} H(x) = 1$ and $\lim_{x \to 0-} H(x) = 0$. In the language introduced earlier in this section we would say that H has a jump (or a jump discontinuity) at the point 0.

The discontinuity can be placed at any point. The function $H(x - c)$ has a jump at $x = c$. Moreover, if $c_1 < c_2 < c_3 < \cdots < c_k$ is a finite sequence of distinct points, then the function

$$F(x) = \sum_{i=1}^{k} H(x - c_i)$$

is a nondecreasing function with jumps at each of the points $c_1, c_2, c_3, \ldots, c_k$. At every other point x_0 it is the case that $\lim_{x \to x_0} F(x) = F(x_0)$.

An interesting question now occurs. We have succeeded in constructing a function that is nondecreasing and has jumps at a prescribed finite set of points. Can we construct such a function if we wish to have jumps at a given infinite set of points? This is a question to which we will return.

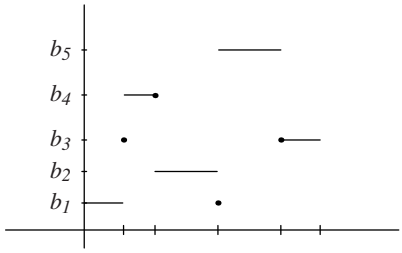


Figure 5.2. Graph of a step function.

Step Functions A function f is a *step function* if it assumes finitely many values, say $b_1, b_2, \ldots, b_N$ and for each $1 \leq i \leq N$ the set

$$f^{-1}(b_i) = \{x : f(x) = b_i\},$$

which represents the set of points at which f assumes the value b_i, is a finite union of intervals and singleton point sets. Another way to think about this is that a function of the form

$$f(x) = \sum_{i=1}^{M} a_i \chi_{A_i}(x)$$

is a step function if all the A_i are intervals or singleton sets. (See Figure 5.2 for an illustration.)

Step functions play an important role in integration theory. They offer a crude way of approximating functions. The function

$$H(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

that we have just seen is a simple step function since it assumes just two values, 0 and 1, where 0 is assumed on the interval $(-\infty, 0)$ and 1 is assumed on $[0, \infty)$.

The discontinuities of a step function are easy to detect.

Distance of a Closed Set to a Point Let C be a closed set and define a function by writing

$$d(x, C) = \inf\{|x - y| : y \in C\}.$$

This function gives a meaning to the distance between a set C and a point x. If $x_0 \in C$, then $d(x_0, C) = 0$, and if $x_0 \notin C$, then $d(x_0, C) > 0$.

This function is continuous at every point; that is, this function has the property that

$$\lim_{x \to x_0} d(x, C) = d(x_0, C). \tag{1}$$

We can interpret (1) geometrically: If two points x_1 and x_2 are close together, then they are at roughly the same distance from the closed set C.

The Characteristic Function of the Cantor Set Let K be the Cantor set and let χ_K be its characteristic function; that is, let $\chi_K = 1$ if $x \in K$ and $\chi_K(x) = 0$ otherwise. This

function has the property that

$$\lim_{x\to x_0} \chi_K(x) = 0$$

if x_0 is not in the Cantor set and the limit exists at no point in the Cantor set. For an easy proof of this you will have to review the properties of the Cantor set and its complement in Exercises 4.3.23 and 4.4.9.

Exercises

5.2.31 Give a proof that includes all necessary details that the limit

$$\lim_{x\to x_0} p(x) = p(x_0)$$

for all polynomials p.

5.2.32 Suppose that you know that

$$\lim_{x\to 2} e^x = e^2.$$

Prove that $\lim_{x\to x_0} e^x = e^{x_0}$ for all x_0.
SEE NOTE 108

5.2.33 Suppose that you know that

$$\lim_{x\to 0} \cos x = 1 \text{ and } \lim_{x\to 0} \sin x = 0.$$

Prove that $\lim_{x\to x_0} \sin x = \sin x_0$ for all x_0.
SEE NOTE 109

5.2.34 In the text we constructed a nondecreasing function with jumps at each of the points c_1, c_2, c_3, …, c_k and continuous everywhere else. Construct an *increasing* function with this property.
SEE NOTE 110

5.2.35 Let $f : [a,b] \to \mathbb{R}$ be a step function. Show that there is a partition

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

so that f is constant on each interval (x_{i-1}, x_i), $i = 1, 2, \ldots, n$.

5.2.36 Suppose that

$$f(x) = \sum_{i=1}^{M} a_i \chi_{A_i}$$

where the A_i are intervals. Show that f is a step function; that is, that f assumes finitely many values, and for each b in the range of f the set $f^{-1}(b)$ is a finite union of intervals or singleton sets. Where are the discontinuities of such a function?
SEE NOTE 111

5.2.37 Show that the characteristic function of the rationals can also be defined by the formula

$$\chi_{\mathbb{Q}}(x) = \lim_{m\to\infty} \lim_{n\to\infty} |\cos(m!\pi x)|^n.$$

5.2.38 Show that

$$\lim_{x\to x_0+} \chi_{\mathbb{Q}}(x) \text{ and } \lim_{x\to x_0-} \chi_{\mathbb{Q}}(x)$$

both fail to exist, where $\chi_{\mathbb{Q}}$ is the characteristic function of the rationals. What would be the answer to the corresponding question for the characteristic function of the irrationals?

5.2.39 Describe the graph of the function $\chi_{\mathbb{Q}}$. What kind of a sketch would convey this set?

5.2.40 Give an example of a set E such that the characteristic function χ_E of E has limits at every point. Can you describe the most general set E with this property?

5.2.41 Give an example of a set E such that the characteristic function χ_E of E has one-sided limits at every point. Can you describe the most general set E with this property?

5.2.42 Show that

$$\lim_{x \to x_0} d(x,C) = d(x_0,C)$$

at every point x_0 where $d(x,C)$ is the distance from x to the closed set C as defined in this section.

5.2.43 Sketch the graph of the function $d(x,C)$ for several closed sets C (e.g., $\{0\}$, $\mathbb{N}$, $[0,1]$, $\{0\} \cup \{1, 1/2, 1/3, 1/4, \dots\}$, and $[0,1] \cup [2,3]$).

5.2.44 Sketch the graph of the characteristic function χ_K of the Cantor set (Exercises 4.3.23 and 4.4.9) and show that

$$\lim_{x \to x_0} \chi_K(x) = 0$$

at all points x not in the Cantor set and that this limit fails to exist at all points in Cantor set.

SEE NOTE 112

5.3 Limits Superior and Inferior

✂ *Advanced section. May be omitted.*

If limits fail to exist we need not abandon all hope of discussing the limiting behavior. We saw this situation in our study of sequence limits in Section 2.13. Even if $\{s_n\}$ diverges so that $\lim_{n\to\infty} s_n$ fails to exist, it is possible that the two extreme limits

$$\liminf_{n\to\infty} s_n \text{ and } \limsup_{n\to\infty} s_n$$

provide some meaningful information. These two concepts always exist (possibly as ∞ or $-\infty$). A similar situation occurs for functions. The theory is nearly identical in many respects.

Definition 5.26: (Lim Sup and Lim Inf) Let $f : E \to \mathbb{R}$ be a function with domain E and suppose that x_0 is a point of accumulation of E. Then we write

$$\limsup_{x \to x_0} f(x) = \inf_{\delta > 0} \sup\{f(x) : x \in (x_0 - \delta, x_0 + \delta) \cap E,\ x \neq x_0\}$$

and

$$\liminf_{x \to x_0} f(x) = \sup_{\delta > 0} \inf\{f(x) : x \in (x_0 - \delta, x_0 + \delta) \cap E,\ x \neq x_0\}$$

As this section is for more advanced readers we have left the development of this concept to the exercises.

Exercises

5.3.1 Show from the definition that

$$\limsup_{x\to x_0} f(x) \geq \liminf_{x\to x_0} f(x).$$

5.3.2 Compute each of the following.

(a) $\limsup_{x\to 0} \sin x^{-1}$

(b) $\limsup_{x\to 0} x \sin x^{-1}$

(c) $\limsup_{x\to 0} x^{-1} \sin x^{-1}$

5.3.3 Formulate an equivalent definition for $\limsup_{x\to 0} f(x)$ expressed in terms of sequential limits; that is, in terms of limits of $f(x_n)$ for $x_n \to x_0$. Show that your definition is equivalent to that in the text.

5.3.4 Give an example of a function f so that

$$\liminf_{x\to 0} f(x) = 0 \text{ and } \limsup_{x\to 0} f(x) = 1.$$

5.3.5 What changes, if any, are there if the definition of lim sup had been written as

$$\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup\{f(x) : x \in (x_0-\delta, x_0+\delta) \cap E\}?$$

SEE NOTE 113

5.3.6 Formulate a definition for the one-sided concepts

$$\limsup_{x\to x_0+} f(x) \text{ and } \limsup_{x\to x_0-} f(x).$$

5.3.7 Give an example of a function f with the properties

$$\liminf_{x\to 0+} f(x) = 0,$$

$$\limsup_{x\to 0+} f(x) = \infty,$$

$$\liminf_{x\to 0-} f(x) = -\infty,$$

and

$$\limsup_{x\to 0-} f(x) = 1.$$

5.3.8 Show that $\lim_{x\to 0} f(x)$ exists if and only if all four of

$$\liminf_{x\to 0+} f(x),$$

$$\limsup_{x\to 0+} f(x),$$

$$\liminf_{x\to 0-} f(x),$$

and

$$\limsup_{x\to 0-} f(x)$$

are equal and finite.

5.3.9 Show that $\lim_{x\to 0} f(x) = \infty$ if and only if

$$\liminf_{x\to 0+} f(x) = \infty,$$
$$\limsup_{x\to 0+} f(x) = \infty,$$
$$\liminf_{x\to 0-} f(x) = \infty,$$

and

$$\limsup_{x\to 0-} f(x) = \infty.$$

5.3.10 For the function $\chi_{\mathbb{Q}}$, the characteristic function of the rationals, determine the values of each of the limits

$$\liminf_{x\to x_0+} \chi_{\mathbb{Q}}(x),\ \limsup_{x\to x_0+} \chi_{\mathbb{Q}}(x),\ \liminf_{x\to x_0-} \chi_{\mathbb{Q}}(x),\ \text{and} \limsup_{x\to x_0-} \chi_{\mathbb{Q}}(x)$$

at any point x_0.

5.3.11 Give an example of a function f such that

$$\{x_0 : \limsup_{x\to x_0-} f(x) > \limsup_{x\to x_0+} f(x)\}$$

is infinite.

5.4 Continuity

The earliest use of the term "continuity" is somewhat clouded by misconceptions of the nature of a function. If a function was given by a single formula then it was considered in the eighteenth century to be "continuous." If, however, the function had a "break" in the formula—defined differently in one interval than in another—it was considered as "discontinuous." As the subject developed these notions continued to obscure the really important ideas. Augustin Cauchy (1789–1857) was the first to give the modern definition and to focus attention on the concept that has now assumed such an important role in analysis.

5.4.1 How to Define Continuity

✂ *Enrichment section. May be omitted.*

Before we proceed to the present day definition, let us consider another notion. Even as late as the middle of the nineteenth century, some mathematicians believed this notion should form the basis for a definition of continuity. This concept is suggested by the phrase "the graph has no jumps." While some instructors of calculus courses might use such phrases to convey a sense of continuity to students, the phrase is not a precise one, nor does it fully convey all we wish a continuous function to be.

This notion *is* related to continuity, however, and has some importance in its own right. We'll begin with a brief discussion of it. Here is one attempt at making our phrase precise. (See Figure 5.3.)

Definition 5.27: (Intermediate Value Property) Let f be defined on an interval I. Suppose that for each $a, b \in I$ with $f(a) \neq f(b)$, and for each d between $f(a)$ and $f(b)$, there exists c between a and b for which $f(c) = d$. We then say that f has the *intermediate value property* (IVP) on I.

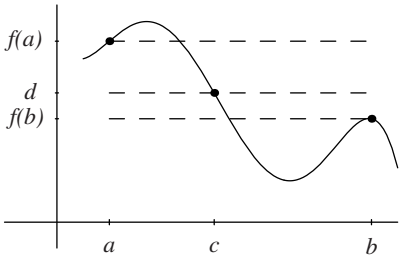


Figure 5.3. At some point between a and b the function assumes any given value d between $f(a)$ and $f(b)$.

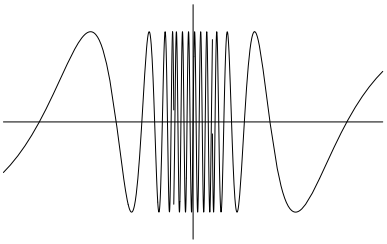


Figure 5.4. Graph of the function $F(x) = \sin x^{-1}$ on $[-\pi/8, \pi/8]$.

Functions with this property are called *Darboux* functions after Jean Gaston Darboux (1842–1917), who showed in 1875 that for every differentiable function F on an interval I, the derivative F' has the IVP on I. He is also particularly famous for his 1875 account of the Riemann integral using upper and lower sums; often reference is made to the "Darboux integral," meaning this version of the classical Riemann integral.

Example 5.28: Let

$$F(x) = \begin{cases} \sin x^{-1} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

The graph of F is shown in Figure 5.4. You may wish to verify that F has the IVP. In particular, F assumes every value in the interval $[-1, 1]$ infinitely often in every neighborhood of $x = 0$. ◀

We haven't yet made precise the phrase "the graph has no jumps," but the IVP seems to convey that idea well enough. Since this property is so easy to describe and appears to have content that is easy to visualize, why *not* take it as the definition of continuity?

Before attempting to answer that question, let us offer a competing phrase to capture the idea of continuity: "If x is near x_0, then $f(x)$ is near $f(x_0)$." As stated, this phrase is not precise, but we can make it precise using the limit concept. This phrase could be interpreted really as asserting that

$$f(x_0) = \lim_{x \to x_0} f(x). \tag{2}$$

According to this criterion our function F of Example 5.28 would *not* be continuous at $x_0 = 0$, because $F(0) = 0$, but $\lim_{x\to 0} F(x)$ does not exist.

We shall see presently that the definition based on limits allows the development of a useful theory. We'll see that the class of continuous functions [as defined using equation (2)] is closed under addition and multiplication, and that such functions have many other desirable properties. For example, the class is closed under certain kinds of limits of sequences, and every continuous function on $[a,b]$ is integrable. On the other hand (as is shown in the exercises), none of the analogous statements is valid for the class of functions defined by IVP.

Thus a theory of continuity based on the limit concept allows a rich structure and enjoys wide applicability, whereas one based on the IVP is rather limited. In addition, the fundamental notion of limit extends to much more general settings than $\mathbb{R}$. In contrast, extensions of IVP, while possible, are peripheral to mathematical analysis.

Exercises

5.4.1 Refer to Example 5.28. Let

$$G(x) = \begin{cases} -F(x) & \text{if } x \neq 0 \\ 1 & \text{if } x = 0. \end{cases}$$

Show that G has the IVP, yet $F+G$ does not. Thus the class of functions with IVP is not closed under addition.

5.4.2 Give an example to show that the class of functions with IVP is not closed under multiplication.

5.4.3 Let

$$H(x) = \begin{cases} x^{-1}\sin x^{-1} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Show that H has IVP on $[0,1]$ but is not bounded there. (This shows that the IVP does not imply boundedness; we shall see that, in contrast, any continuous function on $[0,1]$ would have to be bounded.)

5.4.4 Give an example of a function f with IVP on $[0,1]$ that is bounded but achieves no absolute maximum on [0,1].

5.4.5 Let K be the Cantor set of Exercises 4.3.23 and 4.4.9 and let $\{(a_k,b_k)\}$ be the sequence of intervals complementary to K in (0,1).

(a) Write down a set of equations defining a function f that vanishes at every two-sided point of accumulation of K, is continuous on each interval $[a_k,b_k]$, and for which

$$\lim_{x\to a_k+} f(x) = -1 \text{ and } \lim_{x\to b_k-} f(x) = 1.$$

(See Figure 5.5 for an illustration of one possible choice.)

(b) Verify that f has the intermediate value property.

(c) Verify that f is not continuous in the sense that $f(x_0) = \lim_{x\to x_0} f(x)$ fails at certain points. (Which points?)

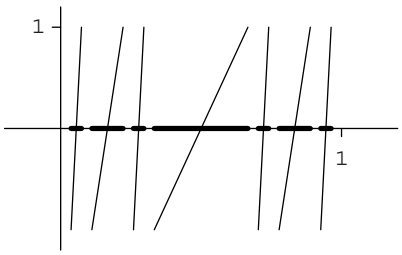


Figure 5.5. Function defined on the complement of the Cantor set, as described in Exercise 5.4.5. The first three stages are shown.

5.4.6 ✂ We construct a function with IVP whose graph may be more difficult to visualize. Let $I_0 = (0,1)$. Each $x \in I_0$ has a unique decimal expansion not ending in a string of 9's. For each $n \in \mathbb{N}$ and $x = .a_1a_2\ldots$ in I_0, let

$$f_n(x) = \frac{a_1(x) + a_2(x) + \cdots + a_n(x)}{n}.$$

Thus $f_n(x)$ represents the average of the first n digits of x. For each $x \in I_0$, let $f(x) = \limsup_n f_n(x)$.

(a) Show that $f : I_0 \to [0,10]$.

(b) Describe how to construct $x \in I_0$ such that $f(x) = \pi$.

(c) Describe how to construct $x \in (.01,.02)$ such that $f(x) = \pi$.

(d) Show that for each interval $(a,b) \subset I_0$ and each $d \in [0,10]$ there exists $c \in (a,b)$ such that $f(c) = d$. Thus, f assumes every value in [0,10] in every interval in I_0. In particular, f has IVP.

(e) Let $A = \{x : f(x) = x\}$. Let $g(x) = 0$ if $x \in A$, $g(x) = f(x)$ for $x \notin A$. Show that $g(x)$ has IVP.

(f) Show that $-g(x) + x$ does not have IVP. Thus the sum of a function with IVP with the identity function need not have IVP.

5.4.2 Continuity at a Point

Let us look at Cauchy's concept of *continuous function*. We begin by defining continuity at a point, more specifically continuity at an interior point of the domain of a function f. This way we are assured that if we are interested in what is happening at the point x_0 then f is defined in a neighborhood of x_0; that is, that f is defined in some interval $(x_0 - c, x_0 + c)$ for a positive number c. This simplifies some of the computations.

Definition 5.29: (Continuous) Let f be defined in a neighborhood of x_0. The function f is *continuous* at x_0 provided $\lim_{x \to x_0} f(x) = f(x_0)$.

This means that for each neighborhood V of $f(x_0)$ there is a neighborhood U of x_0 such that $f(U) \subset V$: that is, if $x \in U$, then $f(x) \in V$. We can, of course, state the definition in terms of δ's and ε's: f is continuous at x_0 if for each $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x) - f(x_0)| < \varepsilon$ whenever $|x - x_0| < \delta$. In the exercises we ask you to verify

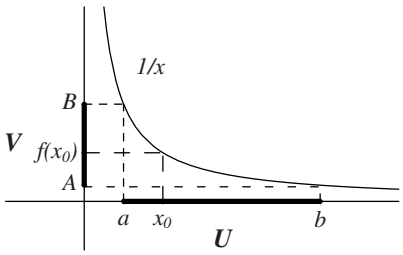


Figure 5.6. Graphical interpretation of the neighborhood definition of continuity for the function $f(x) = x^{-1}$. Note that $a = 1/B$ and $b = 1/A$.

that the three formulations, involving the language of limits, of neighborhoods, and of δ's and ε's, are equivalent. We believe that this is an important exercise for readers who do not yet feel comfortable with the limit concept. Feeling comfortable with the various forms that continuity takes is essential to feeling comfortable with many of the arguments that appear in the sections and chapters that follow.

Observe that a function f can fail to be continuous at x_0 in three ways:

1. f is not defined at x_0.
2. $\lim_{x\to x_0} f(x)$ fails to exist.
3. f is defined at x_0 and $\lim_{x\to x_0} f(x)$ exists, but
$$f(x_0) \neq \lim_{x\to x_0} f(x).$$

We leave it to you to provide simple examples of each of these possibilities.

Example 5.30: Let $f : (0,\infty) \to \mathbb{R}$ be defined by $f(x) = 1/x$. We show that if $x_0 \in (0,\infty)$, then f is continuous at x_0.

Let's first try the "neighborhood" definition of continuity. Let V be a neighborhood of $f(x_0)$, say $V = (A,B)$. Thus $A < f(x_0) < B$. We must find a neighborhood $U = (a,b)$ of x_0 such that $f(U) \subset V$. A picture suggests what to do: Let $a = 1/B$, $b = 1/A$. (See Figure 5.6.) But we must be a bit careful here. Nothing in our neighborhood definition of continuity allows us to assume $A > 0$, so b might not be defined (if $A = 0$), or might not be in the domain of f (if $A < 0$). This presents, however, only a minor nuisance. Thus we assume $A > 0$ in our proof.

So, let us assume $A > 0$, $a = 1/B$, $b = 1/A$. Then, since $A < f(x_0) < B$, we have

$$b = \frac{1}{A} > x_0 = \frac{1}{f(x_0)} > \frac{1}{B} = a,$$

so $x_0 \in (a,b) = U$. Furthermore, if $c \in U$, then $a < c < b$ and

$$B > 1/c = f(c) > A,$$

so $f(c) \in V$. This shows that $f(U) \subset V$ as was required. ◀

Example 5.31: Let us take the same example $f : (0,\infty) \to \mathbb{R}$, defined by $f(x) = 1/x$. We show again that if $x_0 \in (0,\infty)$, then f is continuous at x_0.

We see now how a proof based on the δ-ε definition might look. As with our first proof, we shall provide many details of the proof. After you feel conversant with limits and continuity, you may wish to streamline the proofs somewhat by leaving out details that "any reader finds obvious."

Let $x_0 \in (0,\infty)$ and let $\varepsilon > 0$. We wish to find $\delta > 0$ such that if $|x - x_0| < \delta$ and $x > 0$, then $|1/x - 1/x_0| < \varepsilon$. Rewriting this last inequality as

$$|x - x_0| < \varepsilon x x_0 \tag{3}$$

suggests we try $\delta = \varepsilon x x_0$. But δ should depend only on ε and x_0, not on x. There is no $\delta > 0$ for which the inequality $|x - x_0| < \delta$ implies the inequality

$$|x - x_0| < \varepsilon x x_0$$

for all $x \in (0,\infty)$. We can remove this problem by first requiring x to stay away from 0.

For example, we first require that

$$|x - x_0| < \frac{1}{2} x_0. \tag{4}$$

Then

$$\frac{1}{2} x_0 < x \text{ and} \tag{5}$$

$$\frac{1}{2} \varepsilon x_0^2 < \varepsilon x x_0. \tag{6}$$

The inequalities (3), (4), and (6) suggest taking

$$\delta = \min\left(\frac{1}{2} x_0, \frac{1}{2} x_0^2 \varepsilon\right).$$

For this δ , we compute easily that if $|x - x_0| < \delta$, then

$$\left|\frac{1}{x} - \frac{1}{x_0}\right| = \frac{|x - x_0|}{|x x_0|} < \frac{\frac{1}{2} x_0^2 \varepsilon}{\frac{1}{2} x_0^2} = \varepsilon,$$

the last inequality being obtained by using (6) on the numerator $|x - x_0|$ and (5) on the denominator $|x x_0|$. ◀

Exercises

5.4.7 Prove that the function $f(x) = x^2$ is continuous at every point of $\mathbb{R}$ using the δ-ε form of continuity,

5.4.8 Prove that the function $f(x) = |x|$ is continuous at every point of $\mathbb{R}$ using the δ-ε form of continuity,

5.4.9 Show that the three formulations of continuity appearing at the beginning of this section are equivalent.

5.4.10 In the δ-ε verification of continuity of the function $1/x$ we obtained a δ that did the job. We made no claim that this δ is the largest possible δ we could have chosen. Show that for $\varepsilon = 1$ and $x_0 \in (0,1)$ any δ that works must satisfy $\delta < x_0^2/(1 + x_0)$.

5.4.11 **(Sequence Definition of Continuity)** Prove that f is continuous at x_0 if and only if $\lim_{n\to\infty} f(x_n) = f(x_0)$ for every sequence $\{x_n\} \to x_0$. How would you expect this characterization of continuity at x_0 to be modified if x_0 is not an interior point of its domain?

5.4.12 Give three examples of a function f that fails to be continuous at a point x_0. The first should be discontinuous merely because f is not defined at x_0. The second should be discontinuous because $\lim_{x\to x_0} f(x)$ fails to exist. The third should have neither of these defects but should nonetheless be discontinuous.

5.4.13 A function f is said to be *symmetrically continuous* at a point x if

$$\lim_{h\to 0+} [f(x+h) - f(x-h)] = 0.$$

Show that if f is continuous at a point, then it must be symmetrically continuous there and that the converse does not hold.

5.4.3 Continuity at an Arbitrary Point

To this point we have discussed continuity of a function at an interior point of its domain. How should we modify our notions if x_0 is not an interior point?

Continuity at Endpoints For example, if $f : [a,b] \to \mathbb{R}$, how can we define continuity of f at a or at b? Since the function is defined only on the interval $[a,b]$ and we have defined continuity in terms of limits, it seems that we should require, as before, that for any interior point $x_0 \in (a,b)$

$$\lim_{x\to x_0} f(x) = f(x_0)$$

while at the endpoints continuity would be defined by a one-sided limit,

$$\lim_{x\to a+} f(x) = f(a) \text{ and } \lim_{x\to b-} f(x) = f(b).$$

We can also reformulate our definition in a way that recognizes that f is defined only on $[a,b]$. In our neighborhood definition we interpret U as a *relative* neighborhood of x_0: We require that $f(U \cap [a,b]) \subset V$. Here by a *relative neighborhood* we mean that part of an ordinary neighborhood that is inside the set where f is defined.

Similarly, for the δ-ε definition, our requirement becomes that

$$|f(x) - f(x_0)| < \varepsilon$$

whenever $|x - x_0| < \delta$ and $x \in [a,b]$. Again we are merely restricting our attention to the set where f is defined.

Continuity on an Arbitrary Set These reformulations would work for any set A, not just an interval. Thus we assume that

$$f : A \to \mathbb{R}$$

so that f is a function with domain A and x_0 is an arbitrary point of A, which need not be an interior point nor even a point of accumulation (it might be isolated in A).

There are four versions of the definition. As before, you should check to see that they are indeed equivalent. Some extra care is needed because x_0 could be any point of A and may be isolated in A.

Definition 5.32: (ε-δ Version) Let f be defined on a set A and let x_0 be any point of A. The function f is *continuous* at x_0 provided for every $\varepsilon > 0$ there is a $\delta > 0$ so that

$$|f(x) - f(x_0)| < \varepsilon$$

for every $x \in A$ for which $|x - x_0| < \delta$.

Definition 5.33: (Limit Version) Let f be defined on a set A and let x_0 be any point of A. The function f is *continuous* at x_0 provided either that x_0 is isolated in A or else that x_0 is a point of accumulation of that set and

$$\lim_{x \to x_0} f(x) = f(x_0).$$

Definition 5.34: (Neighborhood Version) Let f be defined on a set A and let x_0 be any point of A. The function f is *continuous* at x_0 provided that for every open set V containing $f(x_0)$ there is an open set U containing x_0 so that $f(U \cap A) \subset V$.

In other words, the neighborhood version asserts that there is a set $U \cap A$ open relative to A that f maps into V. We recall that a set $B \subset A$ is relatively open relative to A if B is the intersection of some open set (here U) with A.

Definition 5.35: (Sequential Version) Let f be defined on a set A and let x_0 be any point of A. The function f is *continuous* at x_0 provided that for every sequence of points $\{x_n\}$ belonging to A and converging to x_0, it follows that $f(x_n) \to f(x_0)$.

Exercises

5.4.14 Prove the equivalence of the four definitions for the continuity of a function defined on an arbitrary set A.

5.4.15 Let $f : \mathbb{N} \to \mathbb{R}$ by writing $f(n) = 1/n^2$. Is f continuous at any point in its domain?
SEE NOTE 114

5.4.16 Using each of the four versions of continuity, show that any function is automatically continuous at any point of its domain that is isolated.

5.4.17 Let f be defined on the set containing the points

$$0,\ 1,\ 1/2,\ 1/4,\ 1/8, \ldots,\ 1/2^n$$

only. What values can you assign at these points that will make this function continuous everywhere where it is defined?
SEE NOTE 115

5.4.18 Let f be defined on the set containing the points $0,\ \pm 1,\ \pm 1/2,\ \pm 1/4,\ \pm 1/8,\ \ldots,\ \pm 1/2^n, \ldots$ only. What values can you assign at these points that will make this function continuous everywhere where it is defined?

5.4.19 If f is continuous at a point x_0 then is it necessarily true that

$$\lim_{x \to x_0} f(x) = f(x_0)?$$

At what points in the domain of f can you say this?
SEE NOTE 116

5.4.20 A function $f : [a,b] \to \mathbb{R}$ is said to be *Lipschitz* if there is a positive number M so that $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in [a,b]$. Show that a Lipschitz function must be continuous. Is the converse true? [Rudolf Otto Sigismund Lipschitz (1832–1903) is probably best remembered for this condition, now forever attached to his name, which he used in formulating an existence theorem for differential equations of the form $y' = f(x,y)$.]

See Note 117

5.4.4 Continuity on a Set

✂ *Enrichment section. May be omitted.*

Continuity is defined at points. A function such as $f(x) = x^2$ could be said to be continuous at every real number x_0, meaning only that $\lim_{x \to x_0} x^2 = x_0^2$ for every real number. In many cases the function considered is continuous at every point in its domain. We say simply that f is continuous. But we must remember this is an assertion about every single point where f is defined.

Definition 5.36: Let $f : A \to \mathbb{R}$. Then f is *continuous* (or *continuous on A*) if f is continuous at each point of A.

If we wish to prove directly from this definition that f is continuous, we must show that f is continuous at every $x_0 \in A$. It is sometimes easier to use the *global* characterization of continuity that follows.

Theorem 5.37: *Let $f : A \to \mathbb{R}$. Then f is continuous if and only if for every open set $V \subset \mathbb{R}$, the set $f^{-1}(V) = \{x \in A : f(x) \in V\}$ is open (relative to A).*

Proof. Suppose first that f is continuous. Let V be open, let $x_0 \in f^{-1}(V)$ and choose $\alpha < \beta$ so that $(\alpha,\beta) \subset V$ and so that $x_0 \in f^{-1}((\alpha,\beta))$. Then $\alpha < f(x_0) < \beta$. We will find a neighborhood U of x_0 such that $\alpha < f(x) < \beta$ for all $x \in U \cap A$. Let $\varepsilon = \min(\beta - f(x_0), f(x_0) - \alpha)$. Since f is continuous at x_0, there exists $\delta > 0$ such that if

$$x \in A \cap (x_0 - \delta, x_0 + \delta),$$

then

$$|f(x) - f(x_0)| < \varepsilon.$$

Thus

$$f(x) - f(x_0) < \beta - f(x_0),$$

and so $f(x) < \beta$. Similarly,

$$f(x) - f(x_0) > \alpha - f(x_0),$$

and so $f(x) > \alpha$. Thus the relative neighborhood $U = (x_0 - \delta, x_0 + \delta) \cap A$ is a subset of $f^{-1}((\alpha,\beta))$ and hence also a subset of $f^{-1}(V)$. We have shown that each member of $f^{-1}(V)$ has a relative neighborhood in $f^{-1}(V)$. That is, $f^{-1}(V)$ is open relative to A.

To prove the converse, suppose f satisfies the condition that for each open interval (α,β) with $\alpha < \beta$, the set $f^{-1}((\alpha,\beta))$ is open relative to A. Let $x_0 \in A$. We must show

that f is continuous at x_0. Let $\varepsilon > 0$, $\beta = f(x_0) + \varepsilon$, $\alpha = f(x_0) - \varepsilon$. Our hypothesis implies that $f^{-1}((\alpha,\beta))$ is open relative to A. Thus

$$f^{-1}((\alpha,\beta)) = \bigcup (a_i, b_i) \cap A,$$

the union being a finite or countable union of pairwise disjoint open intervals. One of these intervals, say (a_j, b_j), contains x_0. Let

$$\delta = \min(x_0 - a_j, b_j - x_0).$$

For $|x - x_0| < \delta$ and $x \in A$ we find

$$\alpha < f(x) < \beta.$$

Because $\beta = f(x_0) + \varepsilon$ and $\alpha = f(x_0) - \varepsilon$ we must have

$$|f(x) - f(x_0)| < \varepsilon.$$

This shows that f is continuous at x_0. ■

We spelled out the details of the proof of Theorem 5.37. This may have caused it to appear rather lengthy. But the proof is nothing more than writing down in a rigorous way what some intuitive pictures indicate. You might find that the neighborhood notion of continuity is a more natural one to use for proving the theorem. We leave this as Exercise 5.4.23.

As a corollary let us point out that we can replace open sets by open intervals; thus to check continuity of a function f it is enough to show that $f^{-1}((\alpha,\beta))$ is open for every interval (α,β).

Corollary 5.38: *Let $f : A \to \mathbb{R}$. Then f is continuous if and only if for every interval (α,β), $f^{-1}((\alpha,\beta))$ is open (relative to A).*

Proof. We verify that the conditions (i) $f^{-1}(V)$ is relatively open for all open $V \subset \mathbb{R}$ and (ii) $f^{-1}((\alpha,\beta))$ is relatively open for all $\alpha < \beta$ are equivalent. But this is immediate. If (i) is satisfied, then (ii) is also, since the requirement (ii) is just a special case of (i). On the other hand, if (ii) is satisfied and

$$V = \bigcup (\alpha_i, \beta_i),$$

then

$$f^{-1}(V) = \bigcup f^{-1}((\alpha_i, \beta_i)).$$

Each of the sets $f^{-1}((\alpha_i,\beta_i))$ is open by hypothesis, so $f^{-1}(V)$ is also open because it is a union of a family of open sets. ■

Example 5.39: Let $f(x) = 1/x$ $(x > 0)$. We find

$$f^{-1}((\alpha,\beta)) = \left(\frac{1}{\beta}, \frac{1}{\alpha}\right).$$

Since $(1/\beta, 1/\alpha)$ is open it would follow that f is continuous on $(0,\infty)$. ◄

Exercises

5.4.21 Prove that the function $f(x) = x^2$ is continuous on $\mathbb{R}$ by using Theorem 5.37.

5.4.22 Prove that the function $f(x) = |x|$ is continuous on $\mathbb{R}$ by using Theorem 5.37.

5.4.23 Prove Theorem 5.37 using the neighborhood definition of continuity.

5.4.24 Let f be continuous in a neighborhood U of the point x_0. If $f(x) < \beta$ for all $x \in U \setminus \{x_0\}$, prove that $f(x_0) \leq \beta$. Show by example that we cannot conclude $f(x_0) < \beta$.

5.4.25 Let f, g be defined on $\mathbb{R}$. Suppose $f(0) = 0$ and f is continuous at $x = 0$. Suppose g is bounded in some neighborhood of zero. Prove that fg is continuous at $x = 0$. Apply this to the function $f(x) = x\sin(1/x)$ $(f(0) = 0)$ at $x = 0$.

5.4.26 Let $x_0 \in \mathbb{R}$. Following are four δ-ε conditions on a function $f : \mathbb{R} \to \mathbb{R}$. Which, if any, of these conditions imply continuity of f at x_0? Which, if any, are implied by continuity at x_0?

(a) For every $\varepsilon > 0$ there exists $\delta > 0$ such that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$.

(b) For every $\varepsilon > 0$ there exists $\delta > 0$ such that if $|f(x) - f(x_0)| < \delta$, then $|x - x_0| < \varepsilon$.

(c) For every $\varepsilon > 0$ there exists $\delta > 0$ such that if $|x - x_0| < \varepsilon$, then $|f(x) - f(x_0)| < \delta$.

(d) For every $\varepsilon > 0$ there exists $\delta > 0$ such that if $|f(x) - f(x_0)| < \varepsilon$, then $|x - x_0| < \delta$.

5.4.27 Let $x_0 \in \mathbb{R}$. Following are four δ-ε conditions on a function $f : \mathbb{R} \to \mathbb{R}$. Which, if any, of these conditions imply continuity of f at x_0? Which, if any, are implied by continuity at x_0?

(a) There exists $\varepsilon > 0$ such that for each $\delta > 0$, if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$.

(b) There exists $\varepsilon > 0$ such that for each $\delta > 0$, if $|f(x) - f(x_0)| < \delta$, then $|x - x_0| < \varepsilon$.

(c) There exists $\varepsilon > 0$ such that for each $\delta > 0$, if $|x - x_0| < \varepsilon$, then $|f(x) - f(x_0)| < \delta$.

(d) There exists $\varepsilon > 0$ such that for each $\delta > 0$, if $|f(x) - f(x_0)| < \varepsilon$, then $|x - x_0| < \delta$.

5.4.28 For each of the eight conditions of Exercises 5.4.26 and 5.4.27, describe in words which functions satisfy the condition. (Some of these conditions characterize familiar classes of functions, including the empty class.)

5.4.29 Let $A \subset \mathbb{R}$, $f : A \to \mathbb{R}$, $g : f(A) \to \mathbb{R}$. Prove that if f is continuous at $x_0 \in A$ and g is continuous at $f(x_0)$, then $g \circ f$ is continuous at x_0. Apply this to prove that if f is continuous at x_0, then $|f|$ is continuous at x_0.

5.4.30 Using the notions of unilateral or one-sided limits, define *left continuity* of a function f at a point x_0. Do the same for *right continuity*. If f is defined in a neighborhood of x_0, prove that f is continuous at x_0 if and only if f is both left continuous and right continuous at x_0.

5.4.31 Let $f : \mathbb{R} \to \mathbb{R}$. Prove that f is continuous if and only if for every closed set $K \subset \mathbb{R}$, the set $f^{-1}(K)$ is closed in $\mathbb{R}$. State carefully and prove the analogous result if $f : A \to \mathbb{R}$, where A is an arbitrary nonempty subset of $\mathbb{R}$.

5.4.32 Suppose f has the IVP on (a, b) and is discontinuous at $x_0 \in (a, b)$. Prove that there exists $y \in \mathbb{R}$ such that $\{x : f(x) = y\}$ is infinite.

5.5 Properties of Continuous Functions

We now present some of the most basic of the properties of continuous functions. The first theorem is an algebraic one; it asserts that the family of continuous functions defined on a set has many of the properties of an *algebra*: elements may be added, subtracted, multiplied, and (under some conditions) divided.

Theorem 5.40: *Let $f, g : A \to \mathbb{R}$ and let $c \in \mathbb{R}$. Suppose f and g are continuous at $x_0 \in A$. Then cf, $f+g$ and fg are continuous at x_0. Furthermore, if $g(x_0) \neq 0$, then f/g is continuous at x_0.*

Proof. The results follow immediately from the limit definition of continuity and the usual algebraic properties of limits. ■

Corollary 5.41: *Every polynomial is continuous on $\mathbb{R}$.*

Proof. The functions $f(x) = 1$ and $g(x) = x$ are continuous on $\mathbb{R}$. The corollary follows from Theorem 5.40. ■

Corollary 5.42: *Every rational function is continuous at each point in its domain (i.e., at each $x \in \mathbb{R}$ at which the denominator does not vanish).*

One of our most important properties allows us to compose two continuous functions. Be careful, though, with the conditions on the domains as they cannot be overlooked.

Theorem 5.43: *Let $f : A \to \mathbb{R}$, $g : B \to \mathbb{R}$ and suppose that $f(A) \subset B$. Suppose that f is continuous at a point $x_0 \in A$ and that g is continuous at the point $y_0 = f(x_0) \in B$. Then the composition function*

$$g \circ f : A \to \mathbb{R}$$

is continuous at x_0.

Proof. This follows from Theorem 5.25. ■

A global version follows as a corollary.

Corollary 5.44: *Let $f : A \to \mathbb{R}$, $g : B \to \mathbb{R}$ and suppose that $f(A) \subset B$. If f is continuous on A and g is continuous on B, then the composition function*

$$g \circ f : A \to \mathbb{R}$$

is continuous on A.

Exercises

5.5.1 If f and g are functions such that $f+g$ is continuous, does it follow that at least one of f or g must be continuous?

5.5.2 If $|f|$ is continuous, does it follow that f is continuous?

5.5.3 If $e^{f(x)}$ is continuous, does it follow that f is continuous?

5.5.4 If $f(f(x))$ is continuous, does it follow that f is continuous?

5.6 Uniform Continuity

Let us take a closer look at the meaning of continuity of a function f on an interval I. The definition asserts that for each $x_0 \in I$ and for every $\varepsilon > 0$, there exists $\delta > 0$ such that if $x \in I$ and $|x - x_0| < \delta$, then

$$|f(x) - f(x_0)| < \varepsilon.$$

Now carefully consider the following statement:

> For every $\varepsilon > 0$, there exists $\delta > 0$ such that if x, $x_0 \in I$ and $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$.

This may appear at first sight to be just a restatement of the meaning of continuity expressed in the first paragraph. If you cannot detect the difference, then you are in good company: Cauchy did not see any difference and used the property just quoted incorrectly to prove that a continuous function on an interval $[a,b]$ must be integrable.

We need to focus on the fact that the number δ depends not only on f and on ε, but also on x_0; that is, $\delta = \delta(f, \varepsilon, x_0)$.

Example 5.45: Consider the function $f(x) = 1/x$ on the interval $I = (0,1)$. We found in Exercise 5.4.10 that if we take $\varepsilon = 1$, we can choose

$$\delta(f, 1, x_0) = \frac{x_0^2}{1 + x_0},$$

but we cannot choose a larger value. Thus if $x_0 \to 0$, then $\delta(f,1,x_0) \to 0$. No number δ is sufficiently small to "work" for *all* $x_0 \in I$. ◀

It is often important to be able to select δ independently of x_0. When this is possible, we say that f is uniformly continuous on I.

Definition 5.46: (Uniformly Continuous) Let f be defined on a set $A \subset \mathbb{R}$. We say that f is *uniformly continuous* (on A) if for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $x, y \in A$ and $|x - y| < \delta$, then $|f(x) - f(y)| < \varepsilon$.

As an illustration of the usefulness of uniform continuity, we note that if f is uniformly continuous on a bounded interval I, then f is bounded on I.

Theorem 5.47: *If a function f is uniformly continuous on a bounded interval I, then f is bounded on I.*

Proof. Here we suppose that I is one of (a,b), $[a,b]$, $[a,b)$, or $(a,b]$. To check that f is bounded, choose δ so that $|f(x) - f(y)| < 1$ whenever $x, y \in I$ and $|x - y| < \delta$. There is a finite set $a = x_0 < x_1 < \cdots < x_n = b$ such that $|x_i - x_{i-1}| < \delta$ for $i = 1, \ldots, n$. Our definition of δ implies that f is bounded on each of the intervals $[x_{i-1}, x_i] \cap I$. Let

$$\begin{aligned} m_i &= \inf\{f(x) : x_{i-1} \le x \le x_i \ , \ x \in I\}, \\ M_i &= \sup\{f(x) : x_{i-1} \le x \le x_i \ , \ x \in I\}, \\ m &= \min\{m_1, \ldots, m_n\} \\ M &= \max\{M_1, \ldots, M_n\}. \end{aligned}$$

Then, for every $x \in I$, $m \leq f(x) \leq M$, so f is bounded on I. ■

Observe that if we tried to present a similar argument for the function $f(x) = 1/x$ on the interval $I = (0,1)$, the continuity of f would allow us to conclude that every $x \in I$ is in an interval on which f is bounded, but we would be unable to obtain a finite number of such intervals that cover I.

In our illustration that uniform continuity on I implies boundedness, we did not specify whether I contained one or more of its endpoints. Our next objective is to show that when $I = [a,b]$ is a *closed* interval, then every function f that is continuous on I is uniformly continuous on I. (Note also the more general version given in Exercise 5.6.14.)

This result will be of importance in many places. In particular, the important result we will later prove, that a continuous function f on $[a,b]$ is integrable, depends on the uniform continuity of f. Cauchy certainly recognized this fact but failed to distinguish between continuity and uniform continuity.

Theorem 5.48: *Let f be continuous on $[a,b]$. Then f is uniformly continuous.*

Proof. Our proof invokes a compactness argument. We recall from our investigations of compactness in Section 4.5 that there are several equivalent formulations possible. We shall use the Bolzano-Weierstrass property. (Exercise 5.6.2 asks for another proof of this same theorem using Cousin's lemma. In Exercise 5.6.13 you are asked to prove it using the Heine-Borel property.)

We use an indirect proof. If f is not uniformly continuous, then there are sequences $\{x_n\}$ and $\{y_n\}$ so that $x_n - y_n \to 0$ but

$$|f(x_n) - f(y_n)| > c$$

for some positive c. (The verification of this step is left as Exercise 5.6.12.)

Now apply the Bolzano-Weierstrass property to obtain a convergent subsequence $\{x_{n_k}\}$. Write z as the limit of this new sequence $\{x_{n_k}\}$. Observe that $x_{n_k} - y_{n_k} \to 0$ since $x_n - y_n \to 0$. Thus $\{x_{n_k}\}$ and the corresponding subsequence $\{y_{n_k}\}$ of the sequence $\{y_n\}$ both converge to the same limit z, which must be a point in the interval $[a,b]$. By the continuity of f, $f(x_{n_k}) \to f(z)$ and $f(y_{n_k}) \to f(z)$. Since $|f(x_n) - f(y_n)| > c$ for all n, this means from our study of sequence limits that

$$|f(z) - f(z)| \geq c > 0$$

and this is impossible. This contradiction proves the theorem. ■

Boundedness of Continuous Functions As an application of Theorem 5.48 we can now prove that any continuous function on a closed bounded interval $[a,b]$ is bounded. Indeed such a function must be uniformly continuous there, and we have already seen in Theorem 5.47 that a uniformly continuous function on a bounded interval is bounded. Thus we have the following useful theorem.

Theorem 5.49: *Let f be continuous on $[a,b]$. Then f is bounded.*

Exercises

5.6.1 Adjust the proof of Theorem 5.48 to show that if f is continuous on a compact set K, then f is uniformly continuous on K.
See Note 118

5.6.2 Give another proof of Theorem 5.48 but this time using Cousin's lemma.
See Note 119

5.6.3 Because of Theorem 5.47 any function that is continuous on $(0,1)$ but unbounded cannot be uniformly continuous there. Give an example of a continuous function on $(0,1)$ that is bounded, but not uniformly continuous.

5.6.4 Let $x_1, x_2, \ldots, x_n$ be real numbers, each in the domain of some function f. Show that f is uniformly continuous on the set $X = \{x_1, x_2, \ldots, x_n\}$.

5.6.5 Let $X = \{x_1, x_2, \ldots, x_n, \ldots\}$. What property must X have so that every function continuous on X is uniformly continuous on X?
See Note 120

5.6.6 Suppose f is uniformly continuous on each of the sets X_1, X_2, ..., X_n and let $X = \bigcup_{i=1}^{n} X_i$. Show that f need not be continuous on X. Show that, even if f is continuous on X, f need not be uniformly continuous on X.

5.6.7 Suppose f is uniformly continuous on each of the compact sets

$$X_1, X_2, \ldots, X_n.$$

Prove that f is uniformly continuous on the set $X = \bigcup_{i=1}^{n} X_i$. Show that this need not be the case if the sets X_k are not closed and need not be the case if the sets X_k are not bounded.
See Note 121

5.6.8 Let f be a uniformly continuous function on a set E. Show that if $\{x_n\}$ is a Cauchy sequence in E then $\{f(x_n)\}$ is a Cauchy sequence in $f(E)$. Show that this need not be true if f is continuous but not uniformly continuous.

5.6.9 A function $f : E \to \mathbb{R}$ is said to be Lipschitz if there is a positive number M so that $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in E$. Show that such a function must be uniformly continuous on E. Is the converse true?
See Note 122

5.6.10 Explain how Exercise 5.6.4 can be deduced from Exercise 5.6.6 or from Exercise 5.6.7.
See Note 123

5.6.11 Give an example of a function f that is continuous on $\mathbb{R}$ and a sequence of compact intervals $X_1, X_2, \ldots, X_n, \ldots$ on each of which f is uniformly continuous, but for which f is not uniformly continuous on $X = \bigcup_{i=1}^{\infty} X_i$.
See Note 124

5.6.12 Show that if f is not uniformly continuous on an interval $[a,b]$ then there are sequences $\{x_n\}$ and $\{y_n\}$ chosen from that interval so that $x_n - y_n \to 0$ but $|f(x_n) - f(y_n)| > c$ for some positive c.
See Note 125

5.6.13 ✂ Prove Theorem 5.48 using the Heine-Borel property.
SEE NOTE 126

5.6.14 Prove the following more general and complete version of Theorem 5.48.

> Suppose that $f : E \to \mathbb{R}$ is continuous. If E is compact, then f must be uniformly continuous on E. Conversely, if every continuous function $f : E \to \mathbb{R}$ is uniformly continuous, then E must be compact.

5.6.15 Prove Theorem 5.49 without using the fact that such a function is uniformly continuous. Use Cousin's lemma.
SEE NOTE 127

5.6.16 Prove Theorem 5.49 without using the fact that such a function is uniformly continuous. Use the Bolzano-Weierstrass property.
SEE NOTE 128

5.6.17 ✂ Prove Theorem 5.49 without using the fact that such a function is uniformly continuous. Use the Heine-Borel property.
SEE NOTE 129

5.7 Extremal Properties

A familiar kind of problem that we study in elementary calculus involves locating extrema of continuous functions defined on an interval $[a,b]$. The technique entails checking values of the function at points where its derivative is zero, at the endpoints of the interval, and at any points of nondifferentiability. For such a process to work, we must be sure the function *has* a maximum (or minimum) on the interval. We verify this now.

Theorem 5.50: *Let f be continuous on $[a,b]$. Then f possesses both an absolute maximum and an absolute minimum.*

Proof. Let $M = \sup\{f(x) : a \le x \le b\}$. By Theorem 5.48, f is uniformly continuous on $[a,b]$. Thus, by Theorem 5.49, $M < \infty$. If there exists x_0 such that $f(x_0) = M$, then f achieves a maximum value M. Suppose, then, that $f(x) < M$ for all $x \in [a,b]$. We show this is impossible.

Let $g(x) = 1/(M - f(x))$. For each $x \in [a,b]$, $f(x) \neq M$; as a consequence, g is continuous and $g(x) > 0$ for all $x \in [a,b]$. From the definition of M we see that

$$\inf\{M - f(x) : x \in [a,b]\} = 0,$$

so

$$\sup\left\{\frac{1}{M - f(x)} : x \in [a,b]\right\} = \infty.$$

This means that g is not bounded on $[a,b]$. This is impossible because, as we saw in Section 5.6, a continuous function defined on a closed interval must be bounded. A similar proof would show that f has an absolute minimum on A. ■

Example 5.51: Does this theorem extend to more general situations? If we replace the interval $[a,b]$ by some other set does the conclusion remain true? The example

$$f(x) = \frac{1}{x} \text{ for } x \in (0,1)$$

shows that the closed interval cannot be replaced by an open one. On the other hand, the example

$$f(x) = x \text{ for } x \in [0,\infty)$$

shows that the bounded closed interval $[a,b]$ cannot be replaced by an unbounded closed one. ◀

From this example the suggestion that we need a closed and bounded set (i.e., a compact set) seems to offer itself. Indeed that is the correct generalization of Theorem 5.50.

Theorem 5.52: *Let f be continuous on a closed and bounded set A. Then f possesses an absolute maximum and an absolute minimum on A.*

Exercises

5.7.1 Give an example of an everywhere discontinuous function that possesses a unique point at which there is an absolute maximum and a unique point at which there is an absolute minimum.

5.7.2 Show that a continuous function maps compact sets to compact sets.
SEE NOTE 130

5.7.3 Prove Theorem 5.50 using a Bolzano-Weierstrass argument.
SEE NOTE 131

5.7.4 Give an example of a function defined only on the rationals and continuous at each point in its domain and yet does not have an absolute maximum.

5.7.5 Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function with the property that

$$\lim_{x\to\infty} f(x) = \lim_{x\to-\infty} f(x) = 0.$$

Show that f has either an absolute maximum or an absolute minimum but not necessarily both.
SEE NOTE 132

5.7.6 Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function that is periodic in the sense that for some number p, $f(x+p) = f(x)$ for all $x \in \mathbb{R}$. Show that f has an absolute maximum and an absolute minimum.

5.8 Darboux Property

We have already observed that the IVP (Darboux property) is not the same as continuity. It is true, however, that if f is continuous on $[a,b]$, then f has the Darboux property. We state Theorem 5.53 in a form that suggests use of Cousin's lemma. (Readers that prefer to use the Bolzano-Weierstrass theorem should see the hint for Exercise 5.8.3.)

Expressed this way the theorem asserts that if the graph has no point on some horizontal line $y = c$, then the graph must be entirely above or below that line. Another way to say this (see Exercise 5.8.8) is that the function must assume every value between any two of its values.

Theorem 5.53: *Let f be continuous on $[a,b]$ and let $c \in \mathbb{R}$. If for every $x \in [a,b]$, $f(x) \neq c$, then either $f(x) > c$ for all $x \in [a,b]$ or $f(x) < c$ for all $x \in [a,b]$.*

Proof. Again, as in the proof of Theorem 5.48, we will use a compactness argument. We shall use Cousin's lemma (Lemma 4.26). In the exercises you are asked to prove this same theorem using the Bolzano-Weierstrass property and the Heine-Borel property.

Let $\mathcal{C}$ denote the collection of closed intervals J such that $f(x) < c$ for all $x \in J$ or $f(x) > c$ for all $x \in J$. We verify that $\mathcal{C}$ forms a Cousin cover of $[a,b]$.

If $x \in [a,b]$, then $|f(x) - c| = \varepsilon > 0$, so there exists $\delta > 0$ such that $|f(t) - f(x)| < \varepsilon$ whenever $|t - x| < \delta$ and $t \in [a,b]$. Thus, if $f(x) < c$, then $f(t) < c$ for all $t \in [x - \delta/2, x + \delta/2]$, while if $f(x) > c$, then $f(t) > c$ for all $t \in [x - \delta/2, x + \delta/2]$. By Cousin's lemma there exists a partition of $[a,b]$, $a = x_0 < x_1 < \cdots < x_n = b$ such that for $i = 1, \ldots, n$, $[x_{i-1}, x_i] \in \mathcal{C}$.

Suppose now that $f(a) < c$. The argument is similar if $f(a) > c$. Since $[a,x_1] = [x_0,x_1] \in \mathcal{C}$, $f(x) < c$ for all $x \in [x_0,x_1]$. Analogously, since $[x_1,x_2] \in \mathcal{C}$, and $f(x_1) < c$, $f(x) < c$ for $x \in [x_1,x_2]$. Proceeding in this way, we see that $f(x) < c$ for all $x \in [a,b]$. ■

You may wish to look at Exercise 5.8.8 for other wordings of this theorem that suggest IVP as "connectedness."

Exercises

5.8.1 Show that a nondecreasing function with the Darboux property must be continuous.

5.8.2 Show that a continuous function maps compact intervals to compact intervals.
SEE NOTE 133

5.8.3 Prove Theorem 5.53 using the Bolzano-Weierstrass property of sequences rather than Cousin's lemma.
SEE NOTE 134

5.8.4 ✂ Prove Theorem 5.53 using the Heine-Borel property.
SEE NOTE 135

5.8.5 Prove Theorem 5.53 using the following "last point" argument: suppose that $f(a) < c < f(b)$ and let z be the last point in $[a,b]$ where $f(z) \leq c$, that is, let

$$z = \sup\{x \in [a,b] : f(x) \leq c\}.$$

Show that $f(z) = c$.
SEE NOTE 136

5.8.6 A function $f : [a,b] \to [a,b]$ is said to have a *fixed point* $c \in [a,b]$ if $f(c) = c$. Show that every continuous function f mapping $[a,b]$ into itself has at least one fixed point.
SEE NOTE 137

5.8.7 Let $f : [a,b] \to [a,b]$ be continuous. Define a sequence recursively by $z_1 = x_1$, $z_n = f(z_{n-1})$ where $x_1 \in [a,b]$. Show that if the sequence $\{z_n\}$ is convergent, then it must converge to a fixed point of f.

5.8.8 Show that Theorem 5.53 can be reworded in the following ways:

(a) Let f be defined and continuous on an interval I, let $a,b \in I$ with $f(a) \neq f(b)$. Let d lie between $f(a)$ and $f(b)$. Then there exists c between a and b such that $f(c) = d$.

(b) A continuous function defined on an interval I maps subintervals of I onto either single points or else subintervals of $\mathbb{R}$. [Singleton points are often considered to be (degenerate) intervals.]

SEE NOTE 138

5.8.9 A continuous function maps compact intervals to compact intervals. Is it true that continuous functions map closed sets to closed sets? Is it true that continuous functions map open sets to open sets?

5.8.10 State forms of Theorem 5.53 and its rewordings in Exercise 5.8.8 for continuous functions defined on intervals that need not be closed and/or bounded.

5.9 Points of Discontinuity

In our discussion of continuous functions we have mentioned discontinuities only as a contrast to the notion of continuity. In many applications of mathematics the functions that arise will have discontinuities and it is well to study such functions. We first ask for a language of discontinuity points. Then we investigate an important class of functions, the monotonic functions, and determine just how badly discontinuous they could be.

5.9.1 Types of Discontinuity

Let x_0 be a point of the domain of some function f. If x_0 is a point of discontinuity, then this means that either the limit $\lim_{x\to x_0} f(x)$ fails to exist or else that limit does exist but

$$f(x_0) \neq \lim_{x\to x_0} f(x).$$

Note that when we discuss discontinuity points we are discussing only points at which the function is defined. (Some calculus texts might call x_0 a point of discontinuity even if $f(x_0)$ fails to be defined. This is not our usage here.)

Note, too, that a discontinuity point cannot occur at an isolated point of the domain of the function.

Removable Discontinuities We can separate these cases into situations of increasing severity. The weakest possibility is that $\lim_{x\to x_0} f(x)$ does indeed exist but fails to equal $f(x_0)$. We call this a *removable discontinuity* of f. The word "removable" suggests that were we merely to assign a new value to $f(x_0)$ we would no longer have a discontinuity.

Jump Discontinuities A little more serious case of discontinuity occurs if the limit

$$\lim_{x\to x_0} f(x)$$

does not exist, but it fails to exist only because

$$\lim_{x\to x_0+} f(x) \text{ and } \lim_{x\to x_0-} f(x),$$

the two one-sided limits, exist but disagree. In that case, no matter what value $f(x_0)$ assumes, this is a point of discontinuity.

We call this a *jump discontinuity* of f. The difference between the two limits

$$\lim_{x\to x_0+} f(x) - \lim_{x\to x_0-} f(x)$$

is a measure of the "size" of the discontinuity and is called the *jump*.

Essential Discontinuities Finally, the most intractable kind of discontinuity would be the situation in which $\lim_{x\to x_0} f(x)$ does not exist, and at least one of the two right-hand and left-hand limits (perhaps both)

$$\lim_{x\to x_0+} f(x) \text{ and } \lim_{x\to x_0-} f(x)$$

also does not exist. Again, no matter what value $f(x_0)$ assumes, this is a point of discontinuity. We call this an *essential discontinuity* of f.

Example 5.54: Let $f(x) = 0$ for all $x \neq 0$ and let $f(0) = 2$. It is clear that 0 is a removable discontinuity of f. Perhaps this example seems entirely artificial. A more natural example would be the function given by the following formula:

$$f(x) = \frac{x+1}{x^2-1} \ (x \neq \pm 1), \ \ f(1) = c_1, \ \ f(-1) = c_2.$$

This function is clearly continuous at every point other than $x = \pm 1$ but may have two discontinuities, one at -1 and one at 1. One of these is not, however, a serious discontinuity since it is removable. You should try to determine which one is removable and which one is essential. ◀

Example 5.55: Let $f(x)$ be defined as the linear function $x+1$ for $x < 0$ and a different linear function $2x-1$ for $x \geq 0$. Then there is a discontinuity at 0 since

$$\lim_{x\to 0+} f(x) = \lim_{x\to 0+} (2x-1) = -1$$

but

$$\lim_{x\to 0-} f(x) = \lim_{x\to 0-} (x+1) = 1.$$

In this case the size of the jump is -2. A picture would show exactly what this jump represents. ◀

Exercises

5.9.1 Show that a function that has the Darboux property cannot have either removable or jump discontinuities.

5.9.2 What kind of discontinuities does the Dirichlet function (see Section 5.2.6) have?

5.9.3 What kind of discontinuities does the characteristic function of the Cantor set (see Section 5.2.6) have?

5.9.4 Let the function $f : \mathbb{R} \to \mathbb{R}$ have just one point of discontinuity and assume only rational values. What kind of discontinuity point must that be?

5.9.5 Classify the discontinuities of the rational function

$$f(x) = \frac{x+1}{x^2-1} \ (x \neq \pm 1), \ \ f(1) = c_1, \ \ f(-1) = c_2.$$

5.9.6 Give an example of a function continuous at 0 but with an essential discontinuity at each other point.

5.9.7 Give an example of a function f with a jump discontinuity and yet $(f)^2$ is continuous everywhere.

5.9.8 Give an example of a function f with an essential discontinuity everywhere and yet $(f)^2$ is continuous everywhere.

5.9.9 Define a function F by the formula

$$F(x) = \lim_{n\to\infty} \frac{x^n}{1+x^n}.$$

What is the domain of this function? Classify all discontinuities.

5.9.2 Monotonic Functions

In general, there is not too much to say about the continuity of an arbitrary function. It is possible for a function to be discontinuous everywhere. But if the function is monotonic this is not possible. We start with some definitions, needed here and again later in many places.

Definition 5.56: (Nondecreasing) Let f be real valued on an interval I. If $f(x_1) \leq f(x_2)$ whenever x_1 and x_2 are points in I with $x_1 < x_2$, we say f is *nondecreasing* on I.

Definition 5.57: (Increasing) Let f be real valued on an interval I. If the strict inequality $f(x_1) < f(x_2)$ holds whenever x_1 and x_2 are points in I with $x_1 < x_2$, we say f is *increasing* on I.

In the opposite direction we define nonincreasing and decreasing.[1]

Definition 5.58: (Nonincreasing) Let f be real valued on an interval I. If $f(x_1) \geq f(x_2)$ whenever x_1 and x_2 are points in I with $x_1 < x_2$, we say f is *nonincreasing* on I.

Definition 5.59: (Decreasing) Let f be real valued on an interval I. If the strict inequality $f(x_1) > f(x_2)$ holds whenever x_1 and x_2 are points in I with $x_1 < x_2$, we say f is *decreasing* on I.

[1] Some authors prefer the terms "increasing" and "strictly increasing" for what we would call nondecreasing and increasing. See the comments on page 298.

A function that is either nonincreasing or nondecreasing is said to be *monotonic*. Sometimes, to emphasize that there is a strict inequality, we say that a function that is increasing or decreasing is *strictly monotonic*.

The class of monotonic functions has a particularly interesting structure as regards continuity. Such functions can never have essential discontinuities. This is because if f is monotonic nondecreasing or monotonic nonincreasing, then at any point both one-sided limits $\lim_{x\to x_0+} f(x)$ and $\lim_{x\to x_0-} f(x)$ exist.

Theorem 5.60: *Let f be monotonic on an interval I. If x_0 is interior to I, then the one-sided limits $\lim_{x\to x_0-} f(x)$ and $\lim_{x\to x_0+} f(x)$ both exist.*

Proof. Suppose f is nondecreasing on I; the proof for the case that f is nonincreasing will then follow by noting that in this case $-f$ is nondecreasing. To prove Theorem 5.60 let x_0 be interior to I and let $\{x_k\}$ be an increasing sequence of points in I such that $\lim_{k\to\infty} x_k = x_0$. Then the sequence $\{f(x_k)\}$ is a nondecreasing sequence of numbers bounded from above by $f(x_0)$. Thus by the monotone convergence principle $\{f(x_k)\}$ approaches a limit L.

For $x_k < x < x_0$,

$$f(x_k) \le f(x) \le L.$$

Let $\varepsilon > 0$. Since $f(x_k) \to L$, there exists $N \in \mathbb{N}$ such that

$$L - f(x_k) < \varepsilon$$

whenever $k \ge N$. For all x satisfying $x_N \le x \le x_0$ we thus have

$$L - f(x) \le L - f(x_N) < \varepsilon.$$

It follows that

$$\lim_{x\to x_0-} f(x) = L,$$

so f has a left-sided limit at x_0. A similar argument shows that f also has a right-sided limit at x_0. ■

Monotonic Functions Have Jump Discontinuities Recall that a function f is said to have a *jump* at x_0 if f has limits from the left and from the right at x_0, but these limits are different. Thus, if f is monotonic nondecreasing, say, then clearly

$$\lim_{x\to x_0-} f(x) \le f(x_0) \le \lim_{x\to x_0+} f(x).$$

Thus the only possibility of a discontinuity at the point x_0 is if the jump

$$J(x_0) = \lim_{x\to x_0+} f(x) - \lim_{x\to x_0-} f(x)$$

is positive. Thus monotonic functions do not have removable discontinuities nor do they have essential discontinuities. They have only jump discontinuities.

Monotonic Functions Have Countably Many Discontinuities We can go further than this. We can ask about the set of points at which there can be a discontinuity point. We ask how large this set can be. The answer is "not very."

Theorem 5.61: *Let f be monotonic on an interval $[a,b]$. Then the set of points of discontinuity of f in that interval is countable. In particular, f must be continuous at the points of a set dense in $[a,b]$.*

Proof. We consider again the case that f is nondecreasing since the case that f is nonincreasing follows by considering the function $-f$. If f is nondecreasing and discontinuous at a point x_0 in (a,b), then the open interval

$$I(x_0) = \left(\lim_{x \to x_0-} f(x), \lim_{x \to x_0+} f(x) \right)$$

either contains no points in the range of f or contains only the single point $f(x_0)$ in the range. (To check this statement, see Exercise 5.9.12.) Thus, each point of discontinuity x_0 of f in I corresponds to an interval $I(x_0)$. For two different points of discontinuity x_1 and x_2, the intervals $I(x_1)$ and $I(x_2)$ are disjoint (because f is nondecreasing). But any collection of disjoint intervals in $\mathbb{R}$ can be arranged into a sequence (Exercise 4.6.10) and so there can be only countably many points of discontinuity of f . ■

It is easy to construct monotonic functions with infinitely many points of discontinuity. For example, if $f(x) = n$ on $[n, n+1)$, then f has jumps at all the integers.

It is natural to ask which countable sets can be the set of discontinuities for some monotonic f. For example, does there exist an increasing function that is discontinuous at every rational number in $\mathbb{R}$? (Exercise 5.9.14 provides an answer.)

Example 5.62: Our theorem shows that a monotonic function has a countable set of points at most where it can be discontinuous. It is easy to find examples of monotonic functions with a prescribed set of discontinuities if the set given to us is finite. Could any countable set be given and we then find a monotonic function that has exactly that set as its points of discontinuity?

The answer, remarkably, is yes. Let C be a countable subset of (a,b). List the elements as $c_1, c_2, c_3, \ldots$. Define the function for $a \leq x \leq b$ as

$$f(x) = \sum_{c_n < x} \frac{1}{2^n}.$$

This function is hard to visualize since it depends on the order of the terms. Clearly, $f(a) = 0$ and $f(b) = 1$. The other values are much less clear. But we can see that there is a jump of magnitude $1/2$ at the point c_1, a jump of magnitude $1/4$ at the point c_2, a jump of magnitude $1/8$ at the point c_3, and so on. The function is strictly increasing on any subinterval in which C is dense and would be constant in any interval that contains no points of C. It can be shown that the only discontinuities occur at the points of C.
◀

Exercises

5.9.10 Construct a function with a jump discontinuity of magnitude -5 at the point $x = 1$ and continuous everywhere else.

5.9.11 Find a monotonic function on $[0,1]$ with discontinuities at $1/3$, $2/3$, and $3/4$ only.

5.9.12 Suppose f is increasing on an interval I. Let x_0 be an interior point of I. Prove that $\lim_{x\to x_0-} f(x) \le f(x_0) \le \lim_{x\to x_0+} f(x)$.

5.9.13 Verify the claims made in Example 5.62 about the function f there.
SEE NOTE 139

5.9.14 Using Example 5.62, show that there is a (strictly) increasing function on $[0,1]$ that is discontinuous at each rational number in $(0,1)$ and continuous at each irrational number.

5.9.15 Show that there is no monotonic function on $[0,1]$ that is discontinuous precisely at each irrational number in $(0,1)$.
SEE NOTE 140

5.9.16 Show that if $f:[a,b]\to\mathbb{R}$ is continuous and increasing, then the inverse function f^{-1} exists and is also continuous and increasing on the interval on which it is defined.
SEE NOTE 141

5.9.17 Let f be a continuous function on an open interval (a,b). Suppose that f has no local maximum or local minimum at any point. Show that f must be monotonic.

5.9.18 Suppose that $f:\mathbb{R}\to\mathbb{R}$ and that $f(x)+\alpha x$ is monotonic for every $\alpha\in\mathbb{R}$. Show that $f(x)=ax+b$ for some a, b.

5.9.19 Let $\{f_n\}$ be a sequence of monotonic functions defined on the interval $[0,1]$. Suppose that

$$f(x)=\lim_{n\to\infty} f_n(x)$$

exists for each $0\le x\le 1$. Show that f is monotonic. (If the word "monotonic" is replaced throughout this problem by "continuous," the exercise would be invalid: show this, too.)

5.9.20 Can the range of an increasing function on the interval $[0,1]$ consist only of rational numbers? Can it consist only of irrational numbers?

5.9.3 How Many Points of Discontinuity?

✂ *Advanced section. May be omitted.*

We have already answered the question as to how many points of discontinuity a monotonic function may have. The set of such points must be countable. We know too that all of these are jump discontinuities; a monotonic function has no removable discontinuities and no essential discontinuities.

What is the situation for an arbitrary function? There are three questions. How many removable discontinuities are possible? How many jump discontinuities are possible? How many essential discontinuities are possible?

Example 5.63: One example that we have seen before shows that there can be a great many essential discontinuities. Let f be the characteristic function of the rational numbers; that is, $f(x)$ is 1 if x is a rational number and is 0 if x is irrational. Clearly,

$$\limsup_{x\to x_0} f(x)=1$$

and

$$\liminf_{x\to x_0} f(x)=0$$

at every point x_0. In particular, the limit does not exist anywhere and so every point is an essential discontinuity. ◀

Surprisingly, though, this is not the case for the removable discontinuities or the jump discontinuities. No function can have an uncountable number of such discontinuities.

Theorem 5.64: *Let f be a real function defined on an interval $[a,b]$. The sets of points in $[a,b]$ at which f has a removable discontinuity and at which f has a jump discontinuity are both countable.*

Proof. Let J be the set of points at which there is a jump discontinuity. Every point of J is in one of the two sets:

$$J_+ = \{x \in (a,b) : \lim_{y\to x+} f(y) > \lim_{y\to x-} f(y)\}$$

or

$$J_- = \{x \in (a,b) : \lim_{y\to x+} f(y) < \lim_{y\to x-} f(y)\}.$$

We shall show that J_+ is countable.

If $x \in J_+$, then

$$\lim_{y\to x+} f(y) > \lim_{y\to x-} f(y)$$

and so there is for any such x at least one rational number r so that

$$\lim_{y\to x+} f(y) > r > \lim_{y\to x-} f(y).$$

Moreover, there then must exist some integer m (depending on x and r) so that

$$f(z) > r > f(y)$$

whenever $x - 1/m < y < x < z < x + 1/m$.

Let J_{rn}, where r is a rational and n a positive integer, denote the set of all points x with the property that $f(y) < r < f(z)$ whenever

$$x - 1/n < y < x < z < x + 1/n.$$

We claim that this set is countable. If not, then it must have a point of accumulation and, in particular, there would have to be at least three points $a < b < c$, with $c - a < 1/n$, all belonging to J_{rn}. But by the way that J_{rn} was defined this means, since a and $c \in J_{rn}$, that $f(b) < r$ and $r < f(b)$ are both true. Since this is impossible, all points in J_{rn} are isolated and hence J_{rn} is countable. The union

$$\bigcup_{r\in\mathbb{Q}} \bigcup_{n=1}^{\infty} J_{rn}$$

is a countable union of countable sets and is thus also countable. But this set contains every point of J_+ and so that set is also countable. Similarly, it is true that J_- is countable and hence the set of points with jump discontinuities is countable.

That the set of points at which the function has a removable discontinuity is also countable is left as an exercise. The ideas of the proof here can be used to prove it in a

similar fashion. Notice especially this technique of inserting a rational number between two unequal numbers. ■

Incidentally, this theorem throws a new light on the theorem about the discontinuity points of monotonic functions. In that proof we used the properties of monotonic functions to show that the collection of discontinuity points was countable. But we know easily that the only such points are the jump discontinuities and any function, monotonic or not, has only countably many of these points by our theorem here. Thus we have another way of looking at Theorem 5.61.

Exercises

5.9.21 Give an example of a function with a dense set of removable discontinuities.

5.9.22 Give an example of a function with a dense set of jump discontinuities.

5.9.23 Prove the remaining statement of Theorem 5.64 that is not proved in the text.

5.10 Challenging Problems for Chapter 5

5.10.1 Suppose that f is a function defined on the real line with the property that $f(x+y) = f(x)+f(y)$ for all x, y. Suppose that f is continuous at 0. Show that f must be continuous everywhere.
SEE NOTE 142

5.10.2 Suppose that f is a function defined on the real line with the property that $f(x+y) = f(x)+f(y)$ for all x, y. Suppose that f is continuous at 0. Show that $f(x) = Cx$ for all x and some number C.
SEE NOTE 143

5.10.3 Suppose that f is a function defined on the real line with the property that $f(x+y) = f(x)f(y)$ for all x, y. Suppose that f is continuous at 0. Show that f must be continuous everywhere.
SEE NOTE 144

5.10.4 Generalize Theorem 5.61 to prove that if a function f (not necessarily monotonic) has left-sided limits and right-sided limits at every point of an open interval I, then f must be continuous except on a countable set.

5.10.5 Determine necessary and sufficient conditions on a pair of sets A and B so that they will have the property that there exists a continuous function $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = 0$ for all $x \in A$ and $f(x) = 1$ for all $x \in B$.
SEE NOTE 145

5.10.6 Let $f : [1,\infty)$ be continuous, positive and increasing with $f(x) \to \infty$ as $x \to \infty$. Show that

$$\sum_{k=1}^{\infty} \frac{1}{f(k)}$$

is convergent if and only if the series

$$\sum_{k=1}^{\infty} \frac{f^{-1}(k)}{k^2}$$

converges (where f^{-1} denotes the inverse function).

5.10.7 **(Extensions of continuous functions)** If $f : A \to \mathbb{R}$, $g : B \to \mathbb{R}$, $A \subset B$, and $f(x) = g(x)$ for all $x \in A$, then the function g is said to be an *extension* of the function f. Prove each of the following:

(a) A function that is continuous on a closed set A can be extended to a function that is continuous on $\mathbb{R}$.

(b) A function that is uniformly continuous on a set A can be extended to a function that is uniformly continuous on $\overline{A}$.

(c) A function that is uniformly continuous on an arbitrary nonempty subset of $\mathbb{R}$ can be extended to a function that is uniformly continuous on all of $\mathbb{R}$.

(d) Give an example of a function f that is continuous on (0,1) but that cannot be extended to a function continuous on [0,1].

5.10.8 ✂ For an arbitrary function $f : \mathbb{R} \to \mathbb{R}$ show that

$$\{x_0 : \limsup_{x \to x_0-} f(x) > \limsup_{x \to x_0+} f(x)\}$$

is countable.

5.10.9 ✂ Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ such that there are infinitely many points x_0 at which either

$$f(x_0) > \limsup_{x \to x_0} f(x) \text{ or } f(x_0) < \liminf_{x \to x_0} f(x).$$

5.10.10 ✂ For an arbitrary function $f : \mathbb{R} \to \mathbb{R}$ show that the set of points x_0 at which $f(x_0)$ does not lie between

$$\liminf_{x \to x_0} f(x) \quad \text{and} \quad \limsup_{x \to x_0} f(x)$$

is countable.

5.10.11 ✂ Let y be a real number or $\pm\infty$ and let $f : E \to \mathbb{R}$ be a function. If there is a sequence $\{x_n\}$ of numbers in E and converging to a point c with $x_n \neq c$ and with $f(x_n) \to y$ then y is called *a cluster value* of f at c. Show that every cluster value at c lies between $\liminf_{x \to c} f(x)$ and $\limsup_{x \to c} f(x)$. Show that both $\liminf_{x \to c} f(x)$ and $\limsup_{x \to c} f(x)$ are themselves cluster values of f at c.

5.10.12 Is there a continuous function $f : \mathbb{R} \to \mathbb{R}$ such that for every real y there are precisely two solutions to the equation $f(x) = y$?

5.10.13 Is there a continuous function $f : \mathbb{R} \to \mathbb{R}$ such that for every real y there are precisely three solutions to the equation $f(x) = y$?

5.10.14 Prove that if $f : \mathbb{R} \to \mathbb{R}$, then the set

$$\{x : f \text{ is right continuous at } x \text{ but not left continuous at } x\}$$

is countable.

SEE NOTE 146

Notes

[89] Exercise 5.1.1. Model your answer after Example 5.2.

[90]Exercise 5.1.2. Consider the cases $a = 0$ and $a \neq 0$ separately. If it is easier for you, break into the three cases $a > 0$, $a < 0$, and $a = 0$.

[91]Exercise 5.1.3. Model your answer after Example 5.3.

[92]Exercise 5.1.4. Consider the cases $x_0 = 0$ and $x_0 \neq 0$ separately. Use the factoring trick in Example 5.3 and the device of restricting x to be close to x_0 by assuming that $|x - x_0| < 1$ at least.

[93]Exercise 5.1.8. Don't forget to exclude $x_0 < 0$ from your answer since it is not a point of accumulation of the domain of this function. Consider the cases $x_0 = 0$ and $x_0 > 0$ separately.

[94]Exercise 5.1.12. If $B \subset A$, then the existence of $\lim_{x\to x_0} g(x)$ can be deduced from the existence of $\lim_{x\to x_0} f(x)$. Can you find other conditions? If x_0 is a point of accumulation of $A \cap B$, then the equality of the two limits can be deduced, assuming that both exist.

[95]Exercise 5.1.16. Either find a single sequence

$$x_n \to 0$$

with $x_n \neq 0$ so that the limit

$$\lim_{n\to\infty} |x_n|/x_n$$

does not exist or else find two such sequences with different limits.

[96]Exercise 5.1.22. You could assume (i) that $L > 0$ or (ii) that $f(x) \geq 0$ for all x in its domain. Then convert to a statement about sequences.

[97]Exercise 5.1.28. At $x_0 \neq 0$ the two one-sided limits are equal. What are they? At $x_0 = 0$ they differ.

[98]Exercise 5.1.29. On one side the limit is zero and on the other the limit fails to exist. (Look ahead to Exercise 5.1.38, where you are asked to show that the limit is ∞ which means that the limit does not exist.) You may use the elementary inequality

$$0 < z < e^z$$

(which is valid for all $z > 0$) in your argument. Consider the sequences $1/n \to 0$ and $-1/n \to 0$.

[99]Exercise 5.1.30. Check the definition: There would be no distinction. The limit

$$\lim_{x\to 0-} \sqrt{x},$$

however, would be meaningless since 0 is not a point of accumulation of the domain of the square root function on the left.

[100]Exercise 5.1.34. Use the definitions in this section as a model. You will need a replacement for the "x_0 is a point of accumulation" of the domain condition. If you cannot think of anything better, then simply use the assumption that f is defined in some interval (a, ∞).

[101]Exercise 5.1.38. On one side at 0 the limit is zero and on the other the limit is ∞. See Exercise 5.1.29.

[102]Exercise 5.2.1. Model your proof after Theorem 2.8 for sequences.

[103]Exercise 5.2.3. If the theorem were false, then in every interval $(x_0 - 1/n, x_0 + 1/n)$ there would be a point x_n for which $|f(x_n)| > n$.

[104]Exercise 5.2.9. If x_0 is not a point of accumulation of

$$\text{dom}(f) \cap \text{dom}(g),$$

then the statement

$$\lim_{x\to x_0} f(x) + g(x) = L$$